

ON MATRIX DIFFERENTIATION IN STATISTICS

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SUMMARY

Matrix differentiation methods used in Statistics to find the derivatives of some well-known scalar and matrix functions, are reviewed. Special attention is given to the differentiation of patterned matrices and the evaluation of Jacobians by using matrix differentiation.

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1. Introduction

In many mathematical problems researchers are faced with the task of maximizing or minimizing a function involving a matrix variable. Originally such problems were handled by re-interpreting the matrix problem in terms of the associated scalar problems. The scalar problems were then solved and the results reassembled into a matrix statement. For instance Frazer, Duncan and Collar (1936) differentiated $y = \underline{x}' A \underline{x}$, where A is symmetric, in this way with respect to the elements of $\underline{x}$, reassembled the results and obtained $\frac{\partial y}{\partial \underline{x}} = 2A\underline{x}$. Turnbull (1927), (1930) and (1931) published papers on the properties of a matrix differential operator $\Omega = \left[\frac{\partial}{\partial x_{ji}} \right]$: $p \times p$ which operates on square matrices and thus had taken the first steps in developing a matrix calculus. Dwyer and MacPhail (1948) however laid the foundation of matrix calculus as it is known today. Of particular interest is a thorough study of matrix calculus by Wroblewski (1963) which was based on the

Dwyer-MacPhail matrix derivatives.

The notation used by Dwyer and MacPhail and extended by Dwyer (1967) was rather difficult to use. Trawinski and Bargmann (1964) and Bargmann (1964) introduced a matrix calculus very close to the Dwyer calculus but notationally much simpler. In this paper we refer to this kind of derivative as the element breakdown method and it is discussed in section 5.1.

MacRae (1974) extended the Bargmann calculus to the differentiation, not only of scalar functions with respect to matrices, and matrix functions with respect to scalar variables, but also to the differentiation of matrix functions with respect to matrices. We refer to this kind of derivative as the Kronecker arrangement method of matrix differentiation and it is discussed in section 5.2.

Neudecker (1969), Tracy and Dwyer (1969), McDonald and Swaminathan (1973) and Bentler and Lee (1975) developed another approach by rewriting the original functions to be differentiated as functions of the argument matrices written in column vector notation. These functions are then differentiated with respect to the transposes of similar column vectors to define a new kind of matrix derivative of a matrix function which we will call the vector rearrangement derivative, and is discussed in section 5.3. This method was inspired by the differentiation of functions of vectors with respect to vectors which is discussed in section 3.

Both the Kronecker arrangement method and the vector rearrangement method require the introduction of some new matrices and operators with special properties. These properties are discussed in section 4.

Although it does not look so at first glance, the vector rearrangement method proved to be a very powerful method for writing the matrix derivative as a simple and recognizable statement in terms of the original matrices. This method was developed by different authors who unfortunately used a wide variety of notation, which is very frustrating for the user.

The purpose of this paper is to review matrix derivatives of scalar functions in particular and to consider the different matrix derivative methods in terms of a common notation. Special attention is given to matrix derivatives of patterned matrices such as symmetric matrices, skew symmetric matrices etc. Some of this work is original although we do not claim originality for derivatives involving particular patterned matrices.

Section 2 deals with matrix derivatives of particular scalar valued functions of matrices and scalar derivatives of particular matrix valued functions. In section 5 we consider the three basic kinds of matrix derivative methods while section 6 deals with derivatives of and with respect to patterned matrices. Section 7 deals with higher matrix derivatives which also involve derivatives of Kronecker products. In section 8 derivatives of partitioned matrices are considered and in section 9 the application of matrix derivatives to evaluate Jacobians of matrix transformations is discussed. In section 10 the different

derivative methods are illustrated by obtaining the maximum likelihood estimates of the parameters of the multivariate normal distribution.

2. The differentiation of specific functions of matrices

The following definition gives the matrix equivalent of a scalar variable:

Definition 2.1 (McDonald and Swaminathan (1973))

An $m \times n$ matrix X is said to be mathematically independent and variable if:

- (i) *no element of X is a constant,*
- (ii) *no two or more elements of X are functionally dependent.*

In practice the most common cases where (i) and/or (ii) are violated are when X is a symmetric, skew-symmetric, triangular, diagonal or a correlation pattern matrix.

If y is a scalar valued function of the matrix X , then y is a function of all the different and variable elements of X . If the derivative of y with respect to a particular element x_{ij} is considered, the derivative is a partial derivative and the matrix of partial

derivatives is denoted as $\frac{\partial y}{\partial X}$ (see definition (2.1.1)).

In fact it became standard practice in matrix differentiation to denote all derivatives involving matrices by partial differentials.

2.1 The matrix derivatives of specific scalar functions of mathematically independent and variable matrices

The matrix derivative of a scalar valued function of the argument matrix is:

Definition 2.1.1 (Dwyer and MacPhail (1948), Bargmann (1964))

If $y = y(X)$ is a scalar valued function of the mathematically independent and variable matrix $X: m \times n$, then $\frac{\partial y}{\partial X}$ is defined as the matrix of derivatives $\frac{\partial y}{\partial x_{ij}} : m \times n$.

Using this notation, the following derivatives of scalar functions of a mathematically independent and variable matrix $X: p \times p$ can be derived. Many of these derivatives can also be found in the books of Anderson (1957), Morrison (1967), Graybill (1969), Bellmann (1970), Press (1972), Rao (1973), Graybill (1976) and Bibby and Toutenburg (1977).

If $|X|$ denotes the determinant of $X: p \times p$, then

$$(2.1.1) \quad \frac{\partial |X|}{\partial X} = \tilde{X}' = |X|X^{-T}$$

and in particular $\frac{\partial \ln |X|}{\partial X} = X^{-T}$ if $|X| \neq 0$, where $\tilde{X}$ denotes the adjoint of X and X^{-T}

the transpose of X^{-1} .

If $\text{tr } X$ denotes the trace of $X: p \times p$, then:

$$(2.1.2) \quad \frac{\partial \text{tr } X}{\partial X} = I_p \text{ and } \frac{\partial \text{tr } AX}{\partial X} = A' \text{ (Schönemann (1965)).}$$

If $\text{tr}_j X$ denotes the j -th elementary symmetric function of X ;

$$(2.1.3) \quad \frac{\partial \text{tr}_j X}{\partial X} = (-1)^{j-1} B_{j-1}(X') \text{ (derived jointly by de Waal (1973) and Nel (1972)), where}$$

$$(2.1.4) \quad B_j(X) = \sum_{i=0}^j (-1)^i X^{j-i} \text{tr}_i X.$$

If $w = \underline{z}' X \underline{y}$, where $\underline{z}: p \times 1$, $X: p \times q$ and $\underline{y}: q \times 1$

$$(2.1.5) \quad \frac{\partial w}{\partial X} = \underline{z} \underline{y}'.$$

If λ_i is a non-repeated latent root of X with left eigenvector $\underline{v}_i$ and right eigenvector $\underline{u}_i$, the derivative of λ_i with respect to X is given by:

$$(2.1.6) \quad \frac{\partial \lambda_i}{\partial X} = \underline{v}_i (\underline{v}_i' \underline{u}_i)^{-1} \underline{u}_i' \text{ (Lancaster (1964), Neudecker (1967)).}$$

This result is also implied in formula (8.3) p. 69 of Wilkinson (1965). Lancaster (1964) also gives the derivative for the case of repeated roots.

Other forms of the derivative (2.1.6) are given by Bargmann and Nel (1974):

$$(2.1.7) \quad \frac{\partial \lambda_i}{\partial X} = \frac{\prod_{j \neq i} (X' - \lambda_j I_p)}{\prod_{j \neq i} (\lambda_i - \lambda_j)},$$

$$(2.1.8) \quad \frac{\partial \lambda_i}{\partial X} = \frac{\sum_{j=0}^{p-1} B_j(X') \lambda_i^{p-j-1}}{\sum_{j=0}^{p-1} (-1)^j (p-j) \lambda_i^{p-j-1} \text{tr}_j X}, \text{ where } B_j(X) \text{ is defined in (2.1.4),}$$

where X is asymmetric.

Sugiura (1973) gave derivative formulae for $\frac{\partial \lambda_i}{\partial x_{gh}}$ as well as higher derivatives at the point where $X = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$.

Although zonal polynomials (James (1964)) are scalar valued functions of symmetric matrices that are frequently used, general formulae for the matrix derivative of the zonal polynomial $C_K(X)$ expressed in terms of these polynomials are not available. How-

ever if $\lambda_1, \dots, \lambda_p$ are the latent roots of $X = X': p \times p$;

$$(2.1.9) \quad \sum_{i=1}^p \frac{\partial}{\partial \lambda_i} \left(\frac{C_K(X)}{C_K(I_p)} \right) = \lim_{\mu \rightarrow \infty} \frac{C_K(X + \mu I_p) - C_K(X)}{\mu C_K(I_p)} \\ = \sum_{i=1}^p \left(\frac{K}{K^{(i)}} \right) \frac{C_{K^{(i)}}(X)}{C_{K^{(i)}}(I_p)},$$

(James and Constantine (1974), Muirhead (1970).)

where $K = (k_1, \dots, k_p)$ is a partition of $k = \sum_{i=1}^p k_i$ into no more than p parts and where $K^{(i)}$ denotes the partition: $K^{(i)} = (k_1, \dots, k_i - 1, \dots, k_p)$. More differential equations of this kind are given in Muirhead (1970), Constantine and Muirhead (1972) and Fujikoshi (1975). Sugiura (1973) gives the following second derivatives:

$$(2.1.10) \quad \frac{\partial^2}{\partial x_{ij}^2} C_K(X) |_{X=\Lambda} = \frac{2}{\lambda_i - \lambda_j} \left(\frac{\partial}{\partial \lambda_i} - \frac{\partial}{\partial \lambda_j} \right) C_K(\Lambda), \quad i \neq j, \\ \frac{\partial^2}{\partial x_{ii}^2} C_K(X) |_{X=\Lambda} = \frac{\partial^2}{\partial \lambda_i^2} C_K(\Lambda),$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$.

Marx and Nel (1976) gave tables for the matrix derivatives $\frac{\partial C_K(X)}{\partial X}$ in terms of the elementary symmetric functions of X for all partitions of $k = 2(1) 10$.

2.2 Derivatives of specific matrix valued functions of mathematically independent and variable matrices with respect to a scalar variable.

Definition 2.1.1 (Dwyer and MacPhail (1948), Bargmann (1964)).

If $Y = Y(x): p \times q$ is a matrix valued function of the scalar variable x , the derivative of the matrix Y with respect to x is defined as: $\frac{\partial Y}{\partial x} = \left[\frac{\partial y_{gh}}{\partial x} \right]: p \times q$. When the scalar is the i, j -th element of a matrix X , this derivative will be denoted according to Bargmann (1964) as $\frac{\partial Y}{\partial x_{ij}}$, and not as $\frac{\partial Y}{\partial \langle X \rangle_{ij}}$, which is the original Dwyer-MacPhail notation.

In this section we consider matrices $Y=Y(X)$, $U=U(X)$ and $V=V(X)$ which are functions of the mathematically independent and variable matrix $X: m \times n$ and which are conformable with respect to the operations considered. The authors Dwyer and MacPhail (1948), Bargmann (1964), Dwyer (1967), Morrison (1967), Tracy and Dwyer

(1969), McDonald and Swaminathan (1973) and Bibby and Toutenburg (1977) used the following derivatives:

$$(2.2.1) \quad \frac{\partial X}{\partial x_{ij}} = J_{ij}:mxn, \quad \frac{\partial X'}{\partial x_{ij}} = J_{ji}:nxm,$$

where $X:mxn$, X' denotes the transpose of X and where J_{ij} denotes an elemental matrix with zeros in all positions but a one in the i, j -th position. The notation J_{ij} (which is closer to Dwyer and MacPhail's $\langle J \rangle_{ij}$) is used instead of Bargmann's E_{ij} , since E is often used in statistics as an error matrix. Notice that $J_{ij} = \underline{1}'_i \underline{1}_j$, where $\underline{1}_i$ is a vector with all elements zero except the i -th which is a one.

If Y is a function of x , where x may be the i, j -th element of X , then:

$$(2.2.2) \quad \text{If } Y = U+V:pxq \text{ then: } \frac{\partial Y}{\partial x} = \frac{\partial U}{\partial x} + \frac{\partial V}{\partial x}:pxq$$

$$(2.2.3) \quad \text{If } Y = UV \text{ where } U:pxr, V:rxq \text{ then:}$$

$$\frac{\partial Y}{\partial x} = \frac{\partial U}{\partial x} V + U \frac{\partial V}{\partial x}:pxq.$$

$$(2.2.4) \quad \text{If } Y:pxp \text{ and } n \text{ a natural number, then}$$

$$\frac{\partial Y^n}{\partial x} = \sum_{s=1}^n Y^{s-1} \frac{\partial Y}{\partial x} Y^{n-s}.$$

$$(2.2.5) \quad \text{If } Y:pxp \text{ is non-singular:}$$

$$\frac{\partial Y^{-1}}{\partial x} = -Y^{-1} \frac{\partial Y}{\partial x} Y^{-1},$$

$$\frac{\partial Y^{-T}}{\partial x} = -Y^{-T} \frac{\partial Y'}{\partial x} Y^{-T}.$$

3. The derivatives of scalar and vector valued functions of vectors

3.1 Definitions:

If $\underline{y}:px1$ is a vector valued function of the mathematically independent and variable vector $\underline{x}:qx1$, then the derivative $\frac{\partial \underline{y}}{\partial \underline{x}}$, is defined as the pxq matrix with typical element:

$$\left(\frac{\partial \underline{y}}{\partial \underline{x}} \right)_{ij} = \frac{\partial y_i}{\partial x_j}, \text{ where } i=1, \dots, p \text{ and } j=1, \dots, q. \text{ The transpose of this matrix is denoted as}$$

$$\frac{\partial \underline{y}'}{\partial \underline{x}} \text{ and has the typical element: } \left(\frac{\partial \underline{y}'}{\partial \underline{x}} \right)_{ji} = \frac{\partial y_i}{\partial x_j}.$$

3.2 Rules for vector derivatives (Bargmann (1964), Neudecker (1969))

From the above definitions it follows that:

$$(3.2.1) \quad \frac{\partial \underline{x}}{\partial \underline{x}'} = I_q.$$

If $\underline{y} : \underline{p} \times \underline{1}$ is a vector valued function of $\underline{z} : \underline{m} \times \underline{1}$ where $\underline{z}$ is a vector valued function of $\underline{x} : \underline{q} \times \underline{1}$ then:

$$(3.2.2) \quad \frac{\partial \underline{y}}{\partial \underline{x}'} = \frac{\partial \underline{y}}{\partial \underline{z}'} \frac{\partial \underline{z}}{\partial \underline{x}'} \quad (\text{Bargmann (1964): the vector chain rule}) \text{ or}$$

$$\frac{\partial \underline{y}'}{\partial \underline{x}} = \frac{\partial \underline{z}'}{\partial \underline{x}} \frac{\partial \underline{y}'}{\partial \underline{z}}, \text{ as used by McDonald and Swaminathan (1973).}$$

It should be noted that in (3.2.2) the derivative $\frac{\partial \underline{y}}{\partial \underline{z}'}$ is taken as if all elements of $\underline{z}$ are different, i.e. as if $\underline{z}$ is mathematically independent and variable. In the derivative $\frac{\partial \underline{z}}{\partial \underline{x}'}$ care should be taken of repeated and or constant elements of $\underline{z}$ (see section 6.2). The vector $\underline{x}$ is considered to be mathematically independent and variable.

If $p=1$, (3.2.2) also serves as a chain rule for a scalar function y of $\underline{z}$ e.g.:

$$(3.2.3) \quad \frac{\partial y}{\partial \underline{x}'} = \frac{\partial y}{\partial \underline{z}'} \frac{\partial \underline{z}}{\partial \underline{x}'},$$

where the derivative $\frac{\partial y}{\partial \underline{x}'}$ is known as the *gradient* of y .

If $y = \frac{u}{v}$ where u and v are scalar functions of the vector variable $\underline{x}$, we obtain the rule for derivatives of ratios as:

$$(3.2.4) \quad \frac{\partial y}{\partial \underline{x}} = \frac{1}{v} \left(\frac{\partial u}{\partial \underline{x}} - y \frac{\partial v}{\partial \underline{x}} \right).$$

If $\underline{y} = \underline{A} \underline{x}$ where $\underline{A} : \underline{p} \times \underline{q}$, $\underline{y} : \underline{p} \times \underline{1}$ and $\underline{x} : \underline{q} \times \underline{1}$ then:

$$(3.2.5) \quad \frac{\partial \underline{y}}{\partial \underline{x}'} = \underline{A} \text{ and } \frac{\partial \underline{y}'}{\partial \underline{x}} = \underline{A}'.$$

If $\underline{x}, \underline{y} : \underline{p} \times \underline{1}$ with $\frac{\partial \underline{y}}{\partial \underline{x}'}$ nonsingular, then:

$$(3.2.6) \quad \frac{\partial \underline{y}}{\partial \underline{x}'} = \left(\frac{\partial \underline{x}}{\partial \underline{y}'} \right)^{-1}.$$

If $\underline{w} = \underline{x}' B \underline{y}$ where $B: q \times p$,

$$(3.2.7) \quad \frac{\partial \underline{w}}{\partial \underline{x}} = B \underline{y} \text{ and } \frac{\partial \underline{w}}{\partial \underline{y}} = B' \underline{x} \text{ (Anderson (1958))}.$$

If $\underline{w} = \underline{x}' D \underline{x}$ where $D: p \times p$ then:

$$(3.2.8) \quad \frac{\partial \underline{w}}{\partial \underline{x}} = (D + D') \underline{x} \\ = 2 D \underline{x} \quad \text{if } D \text{ is symmetric (Anderson (1958))}.$$

If $\underline{w} = \frac{\underline{x}' D \underline{x}}{\underline{x}' E \underline{x}}$ where $D, E: q \times q$ then:

$$(3.2.9) \quad \frac{\partial \underline{w}}{\partial \underline{x}} = \frac{1}{\underline{x}' E \underline{x}} \{ (D + D') \underline{x} - \frac{\underline{x}' D \underline{x}}{\underline{x}' E \underline{x}} (E + E') \underline{x} \} \text{ (Browne (1979))}$$

Derivatives of eigenvectors:

Using the same notation as in (2.1.6), it is of interest to notice that if X is symmetric and $\underline{u}_j = \underline{v}_j$ (see (2.1.6)), the derivative of the eigenvector $\underline{u}_j$ with respect to y , where $X = X(y)$ is a vector valued function of the scalar y , given by:

$$(3.2.10) \quad \frac{\partial \underline{u}_j}{\partial y} = \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \left(\frac{1}{\lambda_j - \lambda_\alpha} \right) (\underline{u}_j' \frac{\partial X}{\partial y} \underline{u}_\alpha - \underline{u}_\alpha' \frac{\partial X}{\partial y} \underline{u}_j) \underline{u}_\alpha$$

(Neudecker (1967(a)), Jöreskog (1977), Jöreskog and Goldberger (1972)).

The corresponding result for asymmetric matrices is implied by Wilkinson (1965) formula 10.2 p. 70. Neudecker (a) used the result to obtain second derivatives of λ_i . Bargmann (1979) also gives second derivatives for latent roots and vectors in a compartmental analysis situation:

Consider the differential equations $\frac{\partial \underline{y}_t}{\partial t} = A \underline{y}_t$ ($\underline{y}_0$ given), with solutions $\underline{y}_{it} = \sum_{j=1}^p u_{ij} e^{\lambda_j t}$, $\sum_{j=1}^p u_{ij} = Y_{i0}$ (given), where the λ_j are the latent roots of A and the $\underline{u}_j$ are right eigenvectors i.e. $A \underline{u}_j = \underline{u}_j \Lambda$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$. The symbol b_{ij} will be used to denote a distinct element of A to be estimated. Compartment j is said to be non-

conservative if $\sum_{i=1}^p a_{ij} \neq 0$ and then $b_{ij} = a_{ij}$. However if compartment j is conservative i.e. if $\sum_{i=1}^p a_{ii} = 0$ then $a_{jj} = -\sum_{i \neq j} a_{ij}$ and for a scalar function ϕ of A : $\frac{\partial \phi}{\partial b_{ij}} = \frac{\partial \phi}{\partial a_{ij}} - \frac{\partial \phi}{\partial a_{jj}}$.

Now:

$$(3.2.11) \quad \frac{\partial \lambda_j}{\partial b_{gh}} = u_{hj} v_{jg}, \text{ if } \sum_g a_{gh} \neq 0 \text{ (see 2.1.6)}$$

$$= u_{hj} v_{jg} - u_{hj} v_{jh} \text{ if } \sum_g a_{gh} = 0$$

where $V = U^{-1}$, and

$$(3.2.12) \quad \frac{\partial u_{ij}}{\partial a_{gh}} = u_{ij} \sum_{\alpha \neq j}^p \frac{u_{h\alpha} v_{jg}}{(\lambda_j - \lambda_\alpha)} + u_{hj} \sum_{\alpha \neq j}^p \frac{u_{i\alpha} v_{\alpha g}}{(\lambda_j - \lambda_\alpha)}$$

and $\frac{\partial u_{ij}}{\partial b_{gh}} = \frac{\partial u_{ij}}{\partial a_{gh}}$, if compartment h is non-conservative,

$$= \frac{\partial u_{ij}}{\partial a_{gh}}, \text{ if compartment } h \text{ is conservative.}$$

$$(3.2.13) \quad \frac{\partial v_{ij}}{\partial a_{gh}} = v_{ig} \sum_{\alpha \neq i}^p \frac{u_{h\alpha} (v_{\alpha j} - v_{ij})}{(\lambda_i - \lambda_\alpha)},$$

with $\frac{\partial v_{ij}}{\partial b_{gh}} = \frac{\partial v_{ij}}{\partial a_{gh}}$, if compartment h is non-conservative,

$$= \frac{\partial u_{ij}}{\partial a_{gh}} - \frac{\partial u_{ij}}{\partial a_{hh}}, \text{ if compartment } h \text{ is conservative.}$$

The second derivatives are given by:

$$(3.2.14) \quad \frac{\partial^2 \lambda_j}{\partial b_{gh} \partial b_{rs}} = \left(\frac{\partial u_{hj}}{\partial b_{rs}} \right) v'_{jg} + u_{hj} \left(\frac{\partial v'_{jg}}{\partial b_{rs}} \right),$$

where $v'_{jg} = v_{jg}$, if compartment h is non-conservative

$$= v_{jg} - v_{jh}, \text{ if compartment } h \text{ is conservative.}$$

$$\begin{aligned}
 (3.2.15) \quad \frac{\partial^2 u_{ij}}{\partial b_{gh} \partial b_{rs}} = & \frac{\partial u_{hj}}{\partial b_{rs}} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \frac{u_{i\alpha} v'_{\alpha g}}{\lambda_j - \lambda_\alpha} + u_{hj} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \left(\frac{\partial u_{i\alpha}}{\partial b_{rs}} \right) v'_{\alpha k} \\
 & + u_{i\alpha} \left(\frac{\partial v'_{\alpha g}}{\partial b_{rs}} \right) \left(\frac{1}{\lambda_j - \lambda_\alpha} \right) - u_{hj} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \frac{u_{i\alpha} v_{\alpha g}}{(\lambda_j - \lambda_\alpha)^2} \left(-\frac{\partial \lambda_j}{\partial b_{rs}} - \frac{\partial \lambda_\alpha}{\partial b_{rs}} \right) \\
 & + \left\{ \left(\frac{\partial u_{ij}}{\partial b_{rs}} \right) v'_{jg} + \left(\frac{\partial v'_{jg}}{\partial b_{rs}} \right) u_{ij} \right\} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \frac{u_{h\alpha}}{(\lambda_j - \lambda_\alpha)} \\
 & + u_{ij} v'_{jg} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \left(\frac{\partial u_{h\alpha}}{\partial b_{rs}} \right) \left(\frac{1}{\lambda_j - \lambda_\alpha} \right) \\
 & - u_{ij} v'_{jg} \sum_{\substack{\alpha=1 \\ \alpha \neq j}}^p \frac{u_{h\alpha}}{(\lambda_j - \lambda_\alpha)^2} \left(\frac{\partial \lambda_j}{\partial b_{rs}} - \frac{\partial \lambda_\alpha}{\partial b_{rs}} \right),
 \end{aligned}$$

where $v'_{\alpha g} = v_{\alpha g}$, if compartment h is non-conservative,

$= v_{\alpha g} - v_{\alpha h}$, if compartment h is conservative.

Before considering chain rules involving scalar and matrix functions of matrices, it is necessary to give attention to some matrix theory recently developed and which occur frequently in matrix derivatives.

4. Useful matrices in matrix calculus:

4.1 The permutation matrix:

The first matrix to be introduced here is known by many names in the literature. We will refer to it as the permutation matrix, which is also the name given to it by its founder Murnaghan (1938). It is also used under this name by Vartak (1955), Hartwig (1972), Barnett (1973), McDonald and Swaminathan (1973), Vetter (1973, 1975), Bentler and Lee (1975), Conlisk (1976), Swain (1977) and Davis (1979). Other names currently in use for this matrix are: *permuted identity matrix* by Tracy and Singh (1972 (a), 1972 (b)), Singh (1972), MacRae (1974) and Balestra (1976); *commutation matrix* by Magnus and Neudecker (1979), *vec-permutation matrix* by Henderson and Searle (1979 (a), 1979 (b)), and *tensor commutator* by Pollock (1979). We define this matrix according to Tracy and Dwyer (1969) and use the notation $I_{(m, n)}$ due to MacRae (1974).

If $A:mxn$ and r is a factor of m and s is a factor of n , then $A_{(r)}:mxn$ denotes the r -rowpermuted matrix formed by stacking every r -th row of A into $A_{(r)}$ starting with the first row, then the second etc. (Tracy and Dwyer (1969)).

Example

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \\ a_{51} & a_{52} \\ a_{61} & a_{62} \end{bmatrix} \quad \text{then } A_{(2)} = \begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \\ a_{51} & a_{52} \\ a_{21} & a_{22} \\ a_{41} & a_{42} \\ a_{61} & a_{62} \end{bmatrix}$$

If the same procedure is applied to every s -th column of A starting with the first etc., then the resulting matrix $A^{(s)}$ is called the s -columnpermuted form of A .

The m -row permuted matrix $(I_m \otimes I_n)_{(m)} = (I_{mn})_{(m)}$ will now be called the permutation matrix and be denoted as $I_{(n, m)}$. Similarly $(I_{mn})_{(n)} = I_{(m, n)}$. Magnus and Neudecker (1979), who have a detailed paper on the properties of $I_{(n, m)}$ (which they denote as K_{mn}), proved that:

$$(4.1.1) \quad I_{(n, m)} = \sum_{i=1}^m \sum_{j=1}^n J_{ij} \otimes J_{ji}, \text{ where } J_{ij}:mxn \text{ and } J_{ji} \text{ are defined in (2.2.1).}$$

Useful properties of the permutation matrix are:

$$(4.1.2) \quad I_{(m, n)}^{-1} = I'_{(m, n)} = I_{(n, m)} \quad (\text{See also Tracy and Singh (1972 (a)) who denoted } I_{(n, m)} \text{ as } I_{(m)}).$$

If $A:mxn$ and $B:pxq$ then:

$$(4.1.3) \quad I_{(m, p)} (A \otimes B) = (A \otimes B)_{(p)}$$

$$(4.1.4) \quad (A \otimes B) I_{(q, n)} = (A \otimes B)^{(q)}$$

$$(4.1.5) \quad A \otimes B = I_{(p, m)} (B \otimes A) I_{(n, q)}$$

which implies:

$$(4.1.6) \quad (A \otimes B)^{(q)} = (A \otimes B) I_{(q, n)} = I_{(p, m)} (B \otimes A) = (B \otimes A)_{(m)},$$

(Murnaghan (1938), Tracy and Dwyer (1969), Tracy and Singh (1972 (a), Bentler and Lee (1975), Magnus and Neudecker (1979)).

4.2 The Vec-operator

If $X:mn$, then $\text{vec}(X)$ or $\underline{x}_c$ denotes the $mn \times 1$ column vector formed by stacking the columns of X in order of appearance into a single column vector. This idea was mentioned by Sylvester (1884) and Aitken (1949) while the vec notation was introduced by Koopmans et al (1950) and also used by Neudecker (1968), (1969), McDonald and Swaminathan (1973), Browne (1974), Bentler and Lee (1975), Balestra (1976), Conlisk (1976), Swain (1977), Searle (1978), Magnus and Neudecker (1979) and Henderson and Searle (1979 (a), (b)). The subscript notation $\underline{x}_c$ was introduced by Tracy and Dwyer (1969). An extensive list of different symbols used in the literature for this vector is given by Henderson and Searle (1979 (a)). For convenience the vector $\text{vec}X'$ will sometimes be denoted as $\underline{x}_r$.

The permutation matrix can be used to write $\text{vec}X'$ in terms of $\text{vec}X$ and vice versa e.g.:

If $X:mn$ then

$$(4.2.1) \quad \text{vec}X = \underline{x}_c = I_{(m,n)} \underline{x}_r = I_{(m,n)} \text{vec}X' = (\text{vec}X')_{(n)}$$

$$(4.2.2) \quad \text{vec}X' = \underline{x}_r = I_{(n,m)} \underline{x}_c = I_{(n,m)} (\text{vec}X) = (\text{vec}X)_{(m)},$$

(Nel, H.M. (1977), Magnus and Neudecker (1979), Henderson and Searle (1979) (a)).

Notice that if $X:pxp$ is symmetric, then $\text{vec}X = \text{vec}X'$ and $\text{vec}X = I_{(p,p)} \text{vec}X$. If $Y = AXB$ then:

$$(4.2.3) \quad \text{vec}Y = (B' \otimes A) \text{vec}X \text{ and } \text{vec}Y' = (A \otimes B') \text{vec}X'. \text{ (Roth (1934)).}$$

If $Y = CX'D$ with $C:pxn$, $X:mn$, $D:mxq$ then

$$(4.2.4) \quad \text{vec}Y = (C \otimes D')_{(q)} \text{vec}X \text{ (see also Tracy and Dwyer (1969)).}$$

This result holds since:

$$\begin{aligned} \text{vec}Y &= (D' \otimes C) \text{vec}X' = (D' \otimes C) I_{(n,m)} \text{vec}X, \text{ from (4.2.2)} \\ &= I_{(p,q)} (C \otimes D') \text{vec}X, \text{ from (4.1.6)} \\ &= (C \otimes D')_{(q)} \text{vec}X, \text{ from (4.1.4).} \end{aligned}$$

If $X, Z: m \times n$, $Y: p \times q$, $B: n \times q$ and $A: m \times p$ then

(4.2.5) $\text{tr}(X'Z) = (\text{vec}'X)(\text{vec}Z) = (\text{vec}'X')(\text{vec}Z')$, which together with (4.2.3), implies:

$$(4.2.6) \quad \text{tr}(X'AYB') = \underline{x}_C' (B \otimes A) \underline{y}_C = \text{vec}'X (B \otimes A) \text{vec}Y, \\ = \underline{x}_T (A \otimes B) \underline{y}_T = \text{vec}'X' (A \otimes B) \text{vec}Y', \quad (\text{Neudecker (1969)}).$$

where $\text{vec}'X = (\text{vec}X)'$.

5. The three basic matrix derivative methods in use to obtain matrix derivatives of scalar and matrix functions of matrices.

5.1 *The element breakdown method:* (Dwyer and MacPhail (1948), Bargmann (1964), Dwyer (1967), Bibby and Toutenburg (1977)).

The general approach is to obtain the derivative with respect to a typical element of the matrix argument by using the results in sections 2.2 and 2.3 together with the chain rules, which will be defined. Rules are given by which it is possible to reconstruct the matrix derivative from the derivative with respect to a typical element.

Chain rules

(1) If $z = z(y)$ is a scalar valued function of a scalar y , where $y = y(X)$ is a scalar valued function of $X: m \times n$ then:

$$(5.1.1) \quad \frac{\partial z}{\partial X} = \frac{\partial z}{\partial y} \frac{\partial y}{\partial X}, \quad (\text{Bargmann (1964)}).$$

(2) If $z = z(Y)$ is a scalar valued function of the matrix $Y: p \times q$ and $Y = Y(X)$ is a matrix valued function of $X: m \times n$, then the derivative of z with respect to the ij -th element of X is given by:

$$(5.1.2) \quad \frac{\partial z}{\partial x_{ij}} = \text{tr} \left\{ \frac{\partial z}{\partial Y} \frac{\partial Y'}{\partial x_{ij}} \right\}, \quad (\text{Bargmann (1964)}). \quad (\text{See also Curtis (1972) p. 236}).$$

It should be noted that similarly as in the vector chain rule (3.2.2), the derivative $\frac{\partial z}{\partial Y}$, within the trace in (5.1.2), should be taken as if Y is mathematically independent and variable. The particular pattern of Y is taken into account when the

second factor $\frac{\partial Y'}{\partial x_{ij}}$ is found. See sections 6.2 and 6.3.

The rules to be used here are:

$$(5.1.3) \quad \text{tr}(AJ_{ij}) = a_{ji} = (A')_{ij}, \text{ the } i, j\text{-th element of } A', \text{ or}$$

$$\text{tr}(AJ_{ji}) = a_{ij} = (A)_{ij}, \text{ the } i, j\text{-th element of } A.$$

The derivative $\frac{\partial z}{\partial X} = \left[\frac{\partial z}{\partial x_{ij}} \right]$ can be reconstructed from (5.1.2) and (5.1.3) and the following useful properties of the trace:

$$(5.1.4) \quad \text{tr} A = \text{tr} A'; \text{tr}(AB) = \text{tr}(BA) = \text{tr}(A'B').$$

Example:

If $z = \text{tr} AX^{-1}$ then a typical element of $\frac{\partial z}{\partial X}$ is:

$$\frac{\partial z}{\partial x_{ij}} = \text{tr} \left\{ \frac{\partial \text{tr} AX^{-1}}{\partial X^{-1}} \frac{\partial X^{-1}}{\partial x_{ij}} \right\}, \text{ from (5.1.2)}$$

$$= \text{tr} \{ A'(-X^{-T} J_{ji} X^{-T}) \}, \text{ from (2.2.2), (3.2.1) and (2.3.5)}$$

$$= -(X^{-T} A' X^{-T})_{ij}, \text{ from (5.1.3)}.$$

Thus

$$\frac{\partial z}{\partial X} = -X^{-T} A' X^{-T}.$$

Higher derivatives can be constructed by using (5.1.2) repeatedly together with the rules:

$$(5.1.5) \quad \text{tr}(AJ_{ij} B J_{kl}) = a_{li} b_{jk}$$

$\text{tr}(A J_{ij} B J_{kl} C J_{mn}) = a_{ni} b_{jk} c_{lm}$, etc. and the recognition rule (Browne (1974)):

$$(5.1.6) \quad (A \otimes B)_{ij, gh} = a_{jh} b_{ig}.$$

Higher derivatives are discussed in detail in section 7.2.

5.2 The Kronecker product arrangement of matrix derivatives

Definition 5.2.1 (MacRae (1974), Balestra (1976)).

If $Y = Y(X):pxq$ is a matrix valued function of $X:mxn$, the derivative of Y with respect to X is defined as the $mp \times nq$ matrix:

$$(5.2.1) \quad \frac{\partial Y}{\partial X} = Y \otimes \frac{\partial}{\partial X} = \left[\frac{\partial y_{ij}}{\partial X} \right],$$

where $\frac{\partial}{\partial X} = \left[\frac{\partial}{\partial x_{ij}} \right]: mxn$ is a matrix of differential operators and $\otimes$ denotes the Kronecker product between Y and $\frac{\partial}{\partial X}$.

The following two derivatives can be regarded as the basic derivatives in the sense of (2.2.1).

If $X:mxn$ is a mathematically independent and variable matrix:

$$(5.2.2) \quad \frac{\partial X}{\partial X} = \begin{bmatrix} J_{11} & \dots & J_{1n} \\ \vdots & & \vdots \\ J_{m1} & \dots & J_{mn} \end{bmatrix}: m^2 \times n^2 = E_{(m,n)}, \text{ (de Waal and Nel (1977))}$$

where $\frac{\partial X}{\partial x_{ij}} = J_{ij}: mxn$ as in (2.2.1),

$$(5.2.3) \quad \frac{\partial X'}{\partial X} = \begin{bmatrix} J_{11} & \dots & J_{m1} \\ \vdots & & \vdots \\ J_{1n} & \dots & J_{mn} \end{bmatrix}: mn \times mn = I_{(m,n)}, \text{ (MacRae (1974))},$$

where $\frac{\partial X'}{\partial x_{ji}} = J_{ji}: nxm$ as in (2.2.1) and $I_{(m,n)}$ as in section 4.1.

The following results (MacRae (1974)) are equivalent to (2.2.2) to (2.2.5) and the matrices involved are of the same order as in the corresponding expressions:

$$(5.2.4) \quad \frac{\partial Y}{\partial X} = \frac{\partial U}{\partial X} + \frac{\partial V}{\partial X}, \text{ if } Y = U + V,$$

$$(5.2.5) \quad \frac{\partial Y}{\partial X} = \frac{\partial U}{\partial X} (V \otimes I_n) + (U \otimes I_m) \frac{\partial V}{\partial X}, \text{ } X:mxn, \text{ if } Y = UV.$$

In particular if $Y = AXB$; $\frac{\partial Y}{\partial X} = (A \otimes I_m) E_{(m,n)} (B \otimes I_n)$ and if

$$Y = CX'D; \frac{\partial Y}{\partial X} = (C \otimes I_m) I_{(m, n)} (D \otimes I_n), \text{ where } X: mxn.$$

$$(5.2.6) \quad \frac{\partial Y^n}{\partial X} = \sum_{s=1}^n (X^{s-1} \otimes I_p) \frac{\partial Y}{\partial X} (X^{n-s} \otimes I_p), X: p \times p$$

$$(5.2.7) \quad \frac{\partial Y^{-1}}{\partial X} = -(Y \otimes I_m)^{-1} \frac{\partial Y}{\partial X} (Y \otimes I_n)^{-1}, X: mxn.$$

This method can also be used to differentiate scalar functions of matrices by using the following definition and scalar chain rule:

Definition 5.2.2 (MacRae (1974))

If $A = \begin{bmatrix} a_{ij} \end{bmatrix} : mxn$ and $B = \begin{bmatrix} B_{ij} \end{bmatrix} : mp \times nq$, where $B_{ij} : pxq$, the star product between A and B is defined as:

$$A * B = \sum_{i=1}^m \sum_{j=1}^n a_{ij} B_{ij}.$$

Notice that $A * B = \text{tr } AB'$ if $B = \begin{bmatrix} b_{ij} \end{bmatrix} mxn$.

Chain rule: (MacRae (1974))

If $z = z(Y)$ is a scalar valued function of $Y: pxq$ and $Y = Y(X)$ is a matrix valued function of $X: mxn$ then:

$$(5.2.8) \quad \frac{\partial z}{\partial X} = \frac{\partial z}{\partial Y} * \frac{\partial Y}{\partial X}, \text{ where } \frac{\partial z}{\partial Y} \text{ is taken as if } Y \text{ is mathematically independent and variable, as in (5.1.2).}$$

In particular $\frac{\partial z}{\partial x_{ij}} = \frac{\partial z}{\partial Y} * \frac{\partial Y}{\partial x_{ij}} = \text{tr} \left[\frac{\partial z}{\partial Y} \frac{\partial Y'}{\partial x_{ij}} \right]$, which is the same as (5.1.2).

To simplify the obtained derivative the following identities, which are equivalent to (5.1.3), are useful:

If $A: mxp$, $C: q \times n$ then:

$$(5.2.9) \quad B' * (C \otimes I_n) E_{(n, m)} (A \otimes I_m) = C' B' A' = (ABC)', \text{ (de Waal and Nel (1977)).}$$

$$(5.2.10) \quad B' * (C \otimes I_m) I_{(m, n)} (A \otimes I_n) = ABC \text{ (MacRae (1974)).}$$

MacRae (1974) gave more identities which enable the user to simplify functions of

matrices involving the star product.

5.3 The vector rearrangement method

This method of matrix differentiation is essentially similar to the differentiation of a vector with respect to a vector, as discussed in section 3, supplemented by the rules for stacking matrices into column vectors, as discussed in section 4.2.

Definition 5.2.1 (Neudecker (1969), Tracy and Dwyer (1969), McDonald and Swaminathan (1973), McDonald (1976), Bentler and Lee (1975).

If $Y:pxq$ is a matrix valued function of the mathematically independent and variable matrix $X:mxn$, the vector rearrangement derivative is defined as the $pq \times mn$ matrix: $\frac{\partial \text{vec} Y}{(\partial \text{vec} X)'} ,$ also denoted as $\frac{\partial \text{vec} Y}{\partial \text{vec}' X}$ or $\frac{\partial \underline{y}_c}{\partial \underline{x}_c'}$.

Tracy and Dwyer (1969) and McDonald and Swaminathan (1973) considered the derivative $\frac{(\partial \text{vec} Y')'}{\partial \text{vec} X'} = \frac{\partial \underline{y}_r'}{\partial \underline{x}_r'}$, while Neudecker (1969) considered the derivative $\frac{\partial \text{vec}' Y}{\partial \text{vec} X} = \frac{\partial \underline{y}_c'}{\partial \underline{x}_c}$.

The derivative $\frac{\partial \text{vec} Y}{\partial \text{vec}' X}$ can be written in terms of these derivatives e.g. $\frac{\partial \text{vec} Y}{\partial \text{vec}' X} = (\frac{\partial \text{vec}' Y}{\partial \text{vec} X})'$ from section 3.1

and since $\frac{\partial \underline{y}_r'}{\partial \underline{x}_r'} = (\frac{\partial \underline{y}_r}{\partial \underline{x}_r'})', \underline{\dot{y}}_r = \text{vec} Y' = I_{(q,p)} \text{vec} Y$, and

$\underline{x}_r' = \text{vec}' X I'_{(n,m)} = \text{vec}' X I_{(m,n)}$ it follows that:

$$\frac{\partial \underline{y}_r'}{\partial \underline{x}_r'} = I_{(q,p)} \frac{\partial \text{vec} Y}{\partial \text{vec}' X} I_{(m,n)} \text{ and consequently:}$$

$$(5.3.1) \quad \frac{\partial \underline{y}_r'}{\partial \underline{x}_r'} = I_{(n,m)} \left(\frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right)' I_{(p,q)}, \text{ and}$$

$$(5.3.2) \quad \frac{\partial \text{vec} Y}{\partial \text{vec}' X} = I_{(p,q)} \left(\frac{\partial \underline{y}_r'}{\partial \underline{x}_r'} \right)' I_{(n,m)}.$$

Derivatives of the kind $\frac{\partial \text{vec} Y'}{\partial \text{vec}' X}$, $\frac{\partial \text{vec}' Y'}{\partial \text{vec} X}$, $\frac{\partial \text{vec} Y}{\partial \text{vec}' X'}$, and $\frac{\partial \text{vec}' Y}{\partial \text{vec} X'}$ also appear in the literature. (Tracy and Dwyer (1969)).

The arrangement $\frac{\partial \text{vec} Y}{\partial \text{vec}' X}$ is preferable since:

- (i) it is written in terms of the simplest form of the vec notation, i.e. in terms of $\underline{y}_c = \text{vec} Y$ and $\underline{x}_c = \text{vec} X$,
- (ii) it gives rise to chain rules that are similar to the usual scalar chain rules,
- (iii) the arrangement of derivatives in $\frac{\partial \text{vec} Y}{\partial \text{vec}' X}$ is the natural arrangement of derivatives as they appear in Jacobians.

The following result, which is a consequence of (3.2.5) can be regarded as the main instrument to obtain matrix derivatives by the vector rearrangement method:

(5.3.2) If $\text{vec} Y = H \text{vec} X$ where $H: pq \times mn$ is a matrix of constants, then

$$\frac{\partial \text{vec} Y}{\partial \text{vec}' X} = \frac{\partial \underline{y}_c}{\partial \underline{x}_c'} = H. \text{ (Neudecker (1969), Tracy and Singh (1972 (a)).)}$$

The following two derivatives are analogous to (2.2.1) and can be obtained by using (5.3.2):

(5.3.3) If $X: m \times n$ then $\frac{\partial \text{vec} X}{\partial \text{vec}' X} = I_{mn}$ and $\frac{\partial \text{vec} X'}{\partial \text{vec}' X} = I_{(n, m)}$.

The first derivative is trivial, the second can be proved as follows: (see also Neudecker (1969)).

If $Y = X'$ then on taking differentials

$$dY = I_n dX' I_m$$

$$\text{thus } d\text{vec} Y = (I_m \otimes I_n) d\text{vec} X' = (I_m \otimes I_n) I_{(n, m)} d\text{vec} X$$

$$\text{and } \frac{\partial \text{vec} Y}{\partial \text{vec}' X} = I_{(n, m)}, \text{ where } d\text{vec} Y = \text{vec}(dY).$$

The next results (similar to McDonald and Swaminathan (1973)) are equivalent to (2.2.2) to (2.2.5) and (5.2.4) to (5.2.7) and the matrices involved are of the same order:

$$(5.3.4) \quad \frac{\partial \text{vec} Y}{\partial \text{vec}' X} = \frac{\partial \text{vec} U}{\partial \text{vec}' X} + \frac{\partial \text{vec} V}{\partial \text{vec}' X} \text{ if } Y = U + V.$$

If $Y = AXB$ then $\frac{\partial \text{vec} Y}{\partial \text{vec}' X} = B' \otimes A$, from (4.2.3) and if $Y = CX'D$ then $\frac{\partial \text{vec} Y}{\partial \text{vec}' X} = (C \otimes D')_{(q)}$, from (4.2.4) (Tracy and Dwyer (1969), Neudecker (1969)).

If $Y = UV$, then on taking differentials: $dY = I_p(dU)V + U(dV)I_q$ and consequently:

$$d\text{vec} Y = (V \otimes I_p) d\text{vec} U + (I_q \otimes U) d\text{vec} V$$

and

$$(5.3.5) \quad \frac{\partial \text{vec} Y}{\partial \text{vec}' X} = (V \otimes I_p) \frac{\partial \text{vec} U}{\partial \text{vec}' X} + (I_q \otimes U) \frac{\partial \text{vec} V}{\partial \text{vec}' X} \quad (\text{see Neudecker (1969), Bentler and Lee (1978)}).$$

$$(5.3.6) \quad \frac{\partial \text{vec}(Y^n)}{\partial \text{vec}' X} = \sum_{s=1}^n ((Y')^{n-s} \otimes Y^{s-1}) \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \quad \text{if } Y: p \times p.$$

$$(5.3.7) \quad \frac{\partial \text{vec} Y^{-1}}{\partial \text{vec}' X} = -(Y^{-T} \otimes Y^{-1}) \frac{\partial \text{vec} Y}{\partial \text{vec}' X}.$$

If $Y = zU$ where $z = z(X): 1 \times 1$, $U = U(X): p \times q$ and $X: m \times n$ then:

$$(5.3.8) \quad \begin{aligned} \frac{\partial \text{vec} Y}{\partial \text{vec}' X} &= (\text{vec} U) \frac{\partial z}{\partial \text{vec}' X} + z \frac{\partial \text{vec} U}{\partial \text{vec}' X} \\ &= (\text{vec} U) \text{vec}' \left(\frac{\partial z}{\partial X} \right) + z \frac{\partial \text{vec} U}{\partial \text{vec}' X}, \end{aligned}$$

(Nel (1972), McDonald and Swaminathan (1973), also implied by Neudecker (1969)).

McDonald (1976) gave formulae which enable the user to write $\frac{\partial \text{vec}' Y'}{\partial \text{vec} X'}: mn \times pq$ in terms of $\frac{\partial Y}{\partial X}: pm \times qn$ (see section 5.2) and vice versa. These relations can be rewritten as relations between $\frac{\partial \text{vec} Y}{\partial \text{vec}' X}$ and $\frac{\partial Y}{\partial X}$ by using (5.3.1) and (5.3.2) e.g.

$$(5.3.9) \quad \begin{aligned} \frac{\partial Y}{\partial X} &= (I_{pm} \otimes (\text{vec } I_n)') (I_p \otimes I_{(n,m)}) \left(\frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right)' I_{(p,q)} \otimes I_n ((\text{vec } I_p) \otimes I_{nq}), \\ \frac{\partial \text{vec} Y}{\partial \text{vec}' X} &= I_{(p,q)} (I_{pq} \otimes \text{vec } I_n) (I_q \otimes \frac{\partial Y}{\partial X} \otimes I_m) ((\text{vec } I_q) \otimes I_{mn}) I_{(n,m)} \end{aligned}$$

The chain rule: (Neudecker (1969), McDonald and Swaminathan (1973), Bentler and Lee (1978)).

If $Z = Z(Y): r \times s$ is a matrix valued function of $Y: p \times q$, where $Y = Y(X)$ is a matrix

valued function of the mathematically independent and variable matrix $X:mxn$, then

$$(5.3.10) \quad \frac{\partial \text{vec} Z}{\partial \text{vec}' X} = \frac{\partial \text{vec} Z}{\partial \text{vec}' Y} \frac{\partial \text{vec} Y}{\partial \text{vec}' X}.$$

This chain rule is a direct consequence of the vector chain rule (3.2.2). Notice that in $\frac{\partial \text{vec} Z}{\partial \text{vec}' Y}$, Y is always treated as if it is mathematically independent and variable, whether or not this is so.

It is also assumed here that X is mathematically independent and variable. If this is not the case, then $\text{vec} X$ should be reduced to contain only the different and variable elements of X . This is discussed in section 6.

A controversy has developed around this matter, (see Bentler and Lee (1975), McDonald (1976), Bentler and Lee (1978)) but should be cleared if the above approach is taken.

If $p=q=1$, then $z=z(X)$ is a scalar and (5.3.10) reduces to a chain rule for a scalar function of a matrix, equivalent to (5.1.2) and (5.2.8).

The vector rearrangement method can be used to differentiate Hadamard products of matrices:

If $Z = Z(X):pq$ and $Y = Y(X):pq$ are matrix valued functions of the mathematically independent and variable matrix $X:mxn$ then if $Z * Y = [z_{ij} y_{ij}]:pq$ denotes the Hadamard product of Z and Y , then:

$$(5.3.11) \quad \frac{\partial \text{vec}(Z * Y)}{\partial \text{vec}' X} = D(Z) \frac{\partial \text{vec} Y}{\partial \text{vec}' X} + D(Y) \frac{\partial \text{vec} Z}{\partial \text{vec}' X}.$$

(Bentler and Lee (1978)),

where $D(Z):pq \times pq = \text{diag}(z_{11}, \dots, z_{1q}, z_{21}, \dots, z_{2q}, \dots, z_{p1}, \dots, z_{pq})$.

If $Y = B:pq$, is a constant matrix, then (5.4.11) reduces to

$$\frac{\partial \text{vec}(Z * B)}{\partial \text{vec}' X} = D(B) \frac{\partial \text{vec} Z}{\partial \text{vec}' X} \quad (\text{McDonald and Swaminathan (1973)}).$$

6. Vector rearrangement derivatives of, and with respect to patterned matrices

6.1 Patterned matrices, transition matrices and pattern matrices

In contrast to mathematically independent and variable matrices, a matrix $X:pxq$ is said to be patterned if X contains $p^* = pq - s - v$ mathematically independent and variable elements, where s denotes the number of elements that are repeated (or are

mathematically dependent) and v denotes the number of elements of X which are constant, (Nel (1979)). We will concentrate here on the case where s only denotes elements which are repeated in the matrix, either as positive or negative, for example like symmetric, skew symmetric and correlation pattern matrices. Under the case where X contains s elements that are mathematically dependent on the other elements, can be classified the orthogonal matrices $X:pxp$ where $s = \frac{1}{2}p(p-1)$ and $p^* = \frac{1}{2}p(p+1)$. (Tumura (1965)).

Example 6.1.1

$$(1) \text{ If } X = \begin{bmatrix} x_{11} & 0 & -x_{11} & x_{14} \\ x_{11} & x_{22} & x_{23} & x_{22} \end{bmatrix}, p^* = 4, s = 2, v = 1.$$

$$(2) \text{ If } X:pxp \text{ is symmetric, } p^* = \frac{1}{2}p(p+1), s = \frac{1}{2}p(p-1), v = 0.$$

$$(3) \text{ If } S:pxp \text{ is skew symmetric or has a correlation pattern,}$$

$$p^* = \frac{1}{2}p(p-1), s = \frac{1}{2}p(p-1), v = p.$$

$$(4) \text{ If } X:pxp \text{ is lower triangular, } p^* = \frac{1}{2}p(p+1), s = 0, v = \frac{1}{2}p(p-1)$$

$$(5) \text{ If } X:pxp \text{ is a diagonal matrix, } p^* = p, s = 0, v = p(p-1).$$

If the patterned matrix $X:pxp$ is stacked columnwise into a single columnvector excluding all constant elements and repeated elements, starting with the first column, then the second etc., the resulting columnvector will be of order $p^* \times 1$ and will be denoted as $\text{vecp}X$ or $\underline{x}_c^*$. The p in $\text{vecp}X$ or star in $\underline{x}_c^*$ refers to the fact that the repeated and constant elements of the patterned matrix are excluded from $\text{vec}X$. Henderson and Searle (1979 (b)) denote $\text{vecp}X$ as $\text{vech}X$ if $X:pxp$ is symmetric while Magnus and Neudecker (1978) use the symbol $v(X)$.

It is necessary to define a general transition matrix $K_{pq}: pq \times p^*$ (transition not in the sense of stochastic processes) to transform $\text{vec}X$ into $\text{vecp}X$ such that:

$$(6.1.1) \quad \text{vecp}X = K'_{pq} \text{vec}X.$$

Definition 6.1.1 (Nel (1979)).

Let i, j denote the positions of the pq elements of X as they appear when packed out in $\text{vec}X$; let g, h denote the position of the p^* mathematically independent and variable elements of X as they appear when packed out as in $\text{vecp}X$. A typical element of K_{pq} is then defined as:

$$\begin{aligned}
 (K_{pq})_{ij, gh} &= \pm \frac{1}{k}, \text{ if } x_{ij} \text{ appears } k \text{ times in } \text{vec} X \text{ as the positive or negative of } \\
 &\quad x_{gh} \text{ in } \text{vecp}' X. \text{ The plus or minus sign is taken depending on} \\
 &\quad \text{whether } x_{ij} \text{ is the positive or negative of } x_{gh}. \\
 &= 0, \text{ if } x_{ij} \text{ is a constant} \\
 &= 0, \text{ otherwise.}
 \end{aligned}$$

Example 6.1.2

If X is patterned as in Example 6.1.1 (1) then:

$$\text{vec}' X = \begin{bmatrix} x_{11}, x_{11}, 0, x_{22}, -x_{11}, x_{23}, x_{14}, x_{22} \end{bmatrix}$$

$$\text{vecp}' X = \begin{bmatrix} x_{11}, x_{22}, x_{23}, x_{14} \end{bmatrix}$$

and

$$K_{24} = \begin{matrix} & \begin{matrix} 11 & 22 & 23 & 14 \end{matrix} \\ \begin{matrix} 11 \\ 21 \\ 12 \\ 22 \\ 13 \\ 23 \\ 14 \\ 24 \end{matrix} & \begin{bmatrix} \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}$$

Transition matrices were defined by Tracy and Singh (1972) (a)) for symmetric matrices. The following authors: Browne (1974), Bentler and Lee (1978), Magnus and Neudecker (1978), Neudecker (1978) and Henderson and Searle (1979) (b)) defined transition matrices for at least one or more of the symmetric, skew-symmetric, lower triangular and diagonal patterned matrices.

In this paper we use notation closely related to Browne's notation for symmetric transition matrices, since the order of the patterned matrix X appears clearly in this notation. Notice that for a given patterned matrix X the transition matrix K_{pq} from

vecX to vecpX is uniquely defined as above. Notice too that it yields the transition matrices for the special cases of X symmetric, skew-symmetric, correlation pattern, lower triangular and diagonal as defined by the above mentioned authors – for example:

If $X: p \times p$ is symmetric then K_{pp} , now simply denoted by K_p , has a typical element:

$$(K_p)_{ij, gh} = \frac{1}{2} (\delta_{ig} \delta_{jh} + \delta_{ih} \delta_{jg}) \quad i \leq p, j \leq p, g \leq h \leq p.$$

where δ_{ij} represents Kronecker's delta. (Browne (1974)).

Inverse transition matrices:

Similar to (6.1.1) a transition matrix $P: p^* \times pq$ can be defined such that:

$$\text{vec}X = P' \text{vecp}X$$

for any patterned matrix X. Since vecpX contains only the mathematically independent and variable elements of X, it is easily seen that P, which is of full row rank, must be a unique matrix.

The transition matrix K_{pq} in definition 6.1.1 is defined such that it is the Moore-Penrose inverse of P e.g.

$$K_{pq} = P'(PP')^{-1}, \text{ which is of full column rank.}$$

Furthermore P is the Moore-Penrose inverse of K_{pq} e.g.

$P = (K'_{pq} K_{pq})^{-1} K'_{pq}$, which we will henceforth denote as K_{pq}^- , following Browne's (1974) notation when X is symmetric. Now:

$$(6.1.2) \quad \text{vec}X = K_{pq}^- \text{vecp}X:$$

We already noticed that $P = K_{pq}^-$ is the unique matrix giving (6.2.1). K_{pq} is also unique in the sense that it is the only transition matrix satisfying (6.1.1), yielding $P = K_{pq}^-$ as its Moore-Penrose inverse and being the Moore-Penrose inverse of K_{pq} .

Example 6.1.3

The Moore-Penrose inverse of K_{24} in Example 6.1.2 is:

$$K_{24}^- = \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\text{Notice that } (K_{24}^- K_{24}')^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \text{ hence } K_{24}' (K_{24}^- K_{24}')^{-1} = K_{24}^-.$$

Furthermore K_{24} and K_{24}^- respectively satisfy equations (6.1.1) and (6.1.2). (See also Example 6.3.1.)

For certain patterned matrices, K_{pq}^- is the transpose of K_{pq}' e.g.:

If K_p denotes the lower triangular transition matrix $K_p(q) : p^2 \times p(p+1)$ or diagonal transition matrix $K_p(d) : p^2 \times p$, then $K_p^- = K_p'$.

The pattern matrix (Nel (1979))

Any patterned matrix $X : p \times q$ has associated with its particular pattern, a pattern matrix $M_{pq} : p \times q$ defined by : $M_{pq} = K_{pq}^- K_{pq}'$ where K_{pq}^- is the Moore-Penrose inverse of K_{pq} . The thus defined pattern matrix has the following properties.

(a) M_{pq} is symmetric:

$$M_{pq}' = K_{pq}' K_{pq}' = K_{pq}' (K_{pq}' K_{pq})^{-1} K_{pq}' = K_{pq} K_{pq}^- = M_{pq}$$

(b) M_{pq} is idempotent:

$$M_{pq}^2 = K_{pq} K_{pq}^- K_{pq} K_{pq}^- = K_{pq} K_{pq}^- = M_{pq} \quad (\text{see also Searle (1971) p. 12})$$

(c) If $X : p \times q$ does have the pattern described by M_{pq} then:

(6.1.3) $M_{pq} \text{vec} X = \text{vec} X$, since:

$$\begin{aligned} M_{pq} \text{vec} X &= K_{pq} K_{pq}^- \text{vec} X = K_{pq} (K_{pq}' K_{pq})^{-1} \text{vec} X = K_{pq}' \text{vec} X \\ &= \text{vec} X. \end{aligned}$$

If $Y:pxq$ does not have the pattern described by M_{pq} and if $M_{pq} \text{vec} Y = \text{vec} Z$ then $Z:pxq$ does have the pattern described by X since:

$$(6.1.4) \quad M_{pq} \text{vec} Z = M_{pq}^2 \text{vec} Y = M_{pq} \text{vec} Y = \text{vec} Z$$

(d) Left multiplication of K_{pq} , and right multiplication of K_{pq}^- by M_{pq} leave K_{pq} and K_{pq}^- invariant:

$$(6.1.5) \quad M_{pq} K_{pq} = K_{pq} K_{pq}^- K_{pq} = K_{pq}; \quad K_{pq}^- M_{pq} = K_{pq}^- K_{pq} K_{pq}^- = K_{pq}^-.$$

If $p=q$ the pattern matrix M_{pp} is denoted as M_p . (See also Browne (1974)).

The pattern matrix for a symmetric matrix, denoted as $M_p(s)$ is of particular interest.

If $Y:pxp$ is asymmetric, then $X = \frac{1}{2}(Y+Y')$ is symmetric and consequently: $\text{vec} X = \frac{1}{2}(\text{vec} Y + \text{vec} Y') = \frac{1}{2}(I_p^2 + I_{(p,p)})\text{vec} Y$.

Now $\frac{1}{2}(I_p^2 + I_{(p,p)}) \text{vec} X = \text{vec} X$ implies: $M_p(s) = \frac{1}{2}(I_p^2 + I_{(p,p)})$ is the symmetric pattern matrix for matrices of order pxp .

Similarly $M_p(ss) = \frac{1}{2}(I_p^2 - I_{(p,p)})$ is the skew-symmetric pattern matrix.

If M_p is a symmetric, skew-symmetric or correlation pattern matrix then for any matrix $A:pxp$:

$$(6.1.6) \quad M_p(A \otimes A) = (A \otimes A) M_p.$$

(See also Browne (1974), for the symmetric case, and Magnus and Neudecker (1978)).

From (6.1.6) it follows that:

$$(6.1.7) \quad (K_p^- (A \otimes A) K_p)^n = K_p^- (A^n \otimes A^n) K_p, \text{ (Magnus and Neudecker (1978))}$$

$$(K_p^- (A \otimes A) K_p)^{-1} = K_p^- (A^{-1} \otimes A^{-1}) K_p, \text{ (Browne (1974), Nel (1978))}$$

$$(K_p^- (A \otimes A) K_p)^{\frac{1}{2}} = K_p^- (A^{\frac{1}{2}} \otimes A^{\frac{1}{2}}) K_p, \text{ (Magnus and Neudecker (1978))}$$

where $A^{\frac{1}{2}}$ denotes the symmetric square root of a positive definite symmetric matrix A .

Interesting relations exist between the permuted identity, transition and pattern matrices for different patterns, of which a few will be mentioned here. If s denotes symmetry, ss skew-symmetry, l lower triangularity, d diagonality and c the correlation pattern, then;

$$\begin{array}{llll}
I_{(p,p)} K_p(c) & = & K_p(c) & M_p(s) M_p(c) = M_p(c) \\
I_{(p,p)} K_p(ss) & = & K_p(ss) & M_p(s) M_p(ss) = M_p(ss) \\
I_{(p,p)} K_p(l) & \neq & K_p(l) & M_p(s) M_p(d) = M_p(d) \\
I_{(p,p)} K_p(d) & = & K_p(d) & \\
\\
M_p(s) K_p(c) & = & K_p(c) & K_p^-(c) M_p(s) = K_p^-(c) \\
M_p(s) K_p(ss) & = & K_p(ss) & K_p^-(ss) M_p(s) = K_p^-(ss) \\
M_p(s) K_p(l) & = & K_p(s) & \text{(see also Neudecker (1978)).} \\
M_p(s) K_p(d) & = & K_p(d) & K_p^-(d) M_p(s) = K_p^-(d) \\
K_p^-(s) K_p(l) & = & I_{\frac{1}{2}p(p+1)} &
\end{array}$$

Notice that Neudecker (1978) used the notation: $K_p \equiv I_{(p,p)}$, $L_p \equiv K_p'(l)$, $D_p \equiv K_p^-(s)$;

Magnus and Neudecker (1978) used:

$D \equiv K_p^-(s)$, $N \equiv M_p(s)$; Browne (1974) used $H_p \equiv K_p(d)$, while Henderson and Searle (1979 (b)) used $H \equiv K_p^{-1}(s)$, $G \equiv K_p'(s)$.

6.2 Derivatives with respect to patterned matrices

The derivative with respect to a patterned matrix that occurs most frequently is the derivative with respect to a symmetric matrix. Many definitions were given to find this kind of derivative which unfortunately gave rise to confusion. By introducing the following concept, the confusion may be eliminated.

The first two basic derivatives that are needed in this case, are the derivatives $\frac{\partial \text{vec} X}{\partial \text{vec}' X}$ and $\frac{\partial \text{vec} X}{\partial \text{vec} X}$. Using the results:

$\text{vec} X = K_{pq}^{-1} \text{vecp} X$, $\text{vecp} X = K_{pq}' \text{vec} X$ for any patterned matrix $X: p \times q$, and (5.3.2) we obtain

$$(6.2.1) \quad \frac{\partial \text{vec} X}{\partial \text{vecp}' X} = K_{pq}^{-1} \quad \text{and} \quad \frac{\partial \text{vecp} X}{\partial \text{vec}' X} = K_{pq}'$$

To obtain the derivative of a matrix or scalar function $Z = Z(Y)$ where $Y = Y(X)$ and X is

a patterned matrix, we *must* differentiate with respect to $\text{vecp}X$, which is mathematically independent and variable, by using the chain rule (5.3.10):

$$(6.2.2) \quad \frac{\partial \text{vec}Z}{\partial \text{vecp}'X} = \frac{\partial \text{vec}Z}{\partial \text{vec}'Y} \frac{\partial \text{vec}Y}{\partial \text{vecp}'X},$$

where the derivative $\frac{\partial \text{vec}Z}{\partial \text{vec}'Y}$ is taken as if Y is mathematically independent and variable.

The above derivative is the kind that is required when maximum likelihood estimates are required i.e. the derivative with respect to the mathematically independent and variable elements of the parameter matrix X .

If the derivative $\frac{\partial \text{vec}Z}{\partial \text{vec}'X}$ is required for some reason, it can be obtained from (6.2.2) by multiplying with $\frac{\partial \text{vecp}X}{\partial \text{vec}'X}$ e.g.

$$(6.2.3) \quad \frac{\partial \text{vec}Z}{\partial \text{vec}'X} = \frac{\partial \text{vec}Z}{\partial \text{vecp}'X} \frac{\partial \text{vecp}X}{\partial \text{vec}'X}.$$

In particular it follows that:

$$(6.2.4) \quad \frac{\partial \text{vec}X}{\partial \text{vec}'X} = \frac{\partial \text{vec}X}{\partial \text{vecp}'X} \frac{\partial \text{vecp}X}{\partial \text{vec}'X} = K_{pq}' K'_{pq} = M'_{pq}$$

which is not unique. (See also Example 6.3.1)

If X is mathematically independent and variable then $K_{pq} = K_{pq}^- = M_{pq} = I_{pq}$ and $\frac{\partial \text{vec}X}{\partial \text{vec}'X} = I_{pq}$ as in (5.3.3). Notice that the nonuniqueness of (6.2.4) is the main reason why so much confusion arose on the matter of derivatives with respect to patterned matrices and particularly symmetric matrices. This problem is discussed in section 6.3.

Furthermore:

$$(6.2.5) \quad \frac{\partial \text{vec}X}{\partial \text{vecp}'X} = \frac{\partial \text{vecp}X}{\partial \text{vec}'X} \frac{\partial \text{vec}X}{\partial \text{vecp}'X} = K'_{pq} K_{pq}^- = I_{pq}^* \text{ as expected, since}$$

$\text{vecp}X$ is mathematically independent and variable.

Certain patterned matrices result from mathematically independent and variable matrices $X:pxq$ by taking the appropriate "part" of the matrix to form the patterned matrix, which will be denoted as $\text{Pat}(X):pxq$.

These patterned matrices are characterized by the fact that the Moore-Penrose inverse of K_{pq} is K'_{pq} .

Example 6.2.1

If $X:pxp$ then $\text{Diag}(X):pxp$ is a diagonal matrix containing the main diagonal of X , (McDonald and Swaminathan (1973)) and similarly $\text{Ltri}(X):pxp$ will denote the lower triangular matrix formed by replacing the elements above the diagonal by zeroes.

The particular patterned matrix $\text{Pat}(X):pxq$, that can be formed from X , has the property

$$(6.2.6) \quad \text{vec}X = \text{vec}(\text{Pat}(X)).$$

Since $\text{Pat}(X)$ is a patterned matrix, it follows from (6.1.1), (6.1.2), (6.1.4), (6.2.6) and (5.3.2) that:

$$(6.2.7) \quad \text{vec}(\text{Pat}(X)) = M_{pq} \text{vec}X, \text{ consequently } \frac{\partial \text{vec}(\text{Pat}(X))}{\partial \text{vec}'X} = M_{pq}$$

$$(6.2.8) \quad \text{vec}X = K_{pq}' \text{vec}X = K_{pq}' \text{vec}(\text{Pat}(X)), \text{ consequently:}$$

$$\frac{\partial \text{vec}X}{\partial \text{vec}'(\text{Pat}(X))} = K_{pq}' = K_{pq}$$

$$(6.2.9) \quad \text{vec}(\text{Pat}(X)) = K_{pq}' \text{vec}'(\text{Pat}(X)), \text{ consequently:}$$

$$\frac{\partial \text{vec}(\text{Pat}(X))}{\partial \text{vec}'(\text{Pat}(X))} = K_{pq}' = K_{pq}$$

$$(6.2.10) \quad \text{vecp}(\text{Pat}(X)) = K_{pq}' \text{vec}(\text{Pat}(X)), \text{ consequently:}$$

$$\frac{\partial \text{vecp}(\text{Pat}(X))}{\partial \text{vec}'(\text{Pat}(X))} = K_{pq}'$$

$$(6.2.11) \quad \text{vecp}(\text{Pat}(X)) = K_{pq}' M_{pq} \text{vec}X = K_{pq}' \text{vec}X, \text{ consequently:}$$

$$\frac{\partial \text{vecp}(\text{Pat}(X))}{\partial \text{vec}'X} = K_{pq}'$$

McDonald and Swaminathan (1973) also considered derivatives of this type, where $\text{Pat}(X) = \text{Diag}(X)$, but do not give explicit importance to pattern and transition matrices in this respect. Notice that contrary to McDonald and Swaminathan (1973), we conclude that on the basis of the absence of an expression like (5.3.2), the derivative $\frac{\partial \text{vec}X}{\partial \text{vec}'(\text{Pat}X)}$ cannot be defined.

6.3 The uniqueness of derivatives with respect to patterned matrices; symmetric matrices in particular.

If $y = y(X)$ is a scalar valued function of the patterned matrix $X:pxq$, then the derivative $\frac{\partial y}{\partial \text{vecp}'X}$ is unique since:

$$(6.3.1) \quad \frac{\partial y}{\partial \text{vecp}'X} = \frac{\partial y}{\partial \text{vec}'X} \frac{\partial \text{vec}X}{\partial \text{vecp}'X} = \text{vec}' \left(\frac{\partial y}{\partial X} \right) K_{pq}^{-1}, \text{ where } \frac{\partial y}{\partial X} \text{ is}$$

taken as if X is mathematically independent and variable (see (5.3.10)) and K_{pq}^{-1} is the unique matrix which transforms $\text{vecp}X$ into $\text{vec}X$ as in (6.2.1).

Now:

The derivative $\frac{\partial \text{vecp}Y}{\partial \text{vecp}'X}$ will also be unique since $\frac{\partial \text{vecp}Y}{\partial \text{vecp}'X} = \left[\frac{\partial y_{ij}}{\partial \text{vecp}'X} \right]$ and each $\frac{\partial y_{ij}}{\partial \text{vecp}'X}$ is unique. This kind of derivative is used extensively in section 9 for the derivation of Jacobians.

It should be pointed out that the matrix K_{pq} can be defined differently from Definition 6.1.1 and still yields: $\text{vecp}X = K_{pq}' \text{vec}X$, but the new K_{pq} will not be the Moore-Penrose inverse of K_{pq}^{-} .

Example 6.3.1

$$\text{If } X:2 \times 2 \text{ is symmetric then } K_2(s) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } K_2^-(s) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \text{ Notice that } K_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} (= K_2(\bar{x}))$$

also possesses the property $K_2' \text{vec}X = \text{vecp}X$, but K_2 is not the Moore-Penrose inverse of $K_2^-(s)$. Thus if M_2 is defined as $K_2 K_2^-(s)$ then M_2 is not the pattern matrix with the properties as described in section 6.1. $M_2(s) = K_2(s) K_2^-(s)$ however, is the unique pattern matrix for symmetric matrices of order 2.

The element breakdown method (see section 5.1) can also be used to obtain derivatives with respect to the mathematically independent and variable elements of the patterned matrix $X:pxq$. If $y = y(X)$ then if x_{ij}^* denotes the common value of all repetitions of the element x_{ij} , where x_{ij} is that element taken into $\text{vecp}X$, then from (5.1.2):

$$(6.3.2) \quad \frac{\partial y}{\partial x_{ij}^*} = \text{tr} \left\{ \frac{\partial y}{\partial X} \frac{\partial X'}{\partial x_{ij}^*} \right\}$$

where $\frac{\partial y}{\partial X}$ within the trace is taken as if X is mathematically independent and variable. Symbolically this may be written as $\frac{\partial y}{\partial \mathbf{X}}$ to indicate that no functional relations are taken into account. (Using this notation the chain rules (5.1.2) and (5.3.10) could have been written respectively as:

$$\frac{\partial z}{\partial x_{ij}} = \text{tr} \left\{ \frac{\partial z}{\partial \Psi} \frac{\partial \Psi'}{\partial x_{ij}} \right\}$$

and

$$\frac{\partial \text{vec} Z}{\partial \text{vec}' X} = \left(\frac{\partial \text{vec} Z}{\partial \text{vec}' \Psi} \frac{\partial \text{vec} \Psi}{\partial \text{vec}' X} \right).$$

In particular if $y = x_{ij}$, the equality of x_{ij} to other elements is disregarded so that $\frac{\partial x_{ij}}{\partial X} = J_{ij}$.

In the derivative $\frac{\partial X}{\partial x_{ij}^*}$ the pattern of X is however taken into account. For example if X is symmetric then:

$$(6.3.3) \quad \frac{\partial X}{\partial x_{ij}^*} = J_{ij} + (1 - \delta_{ij}) J_{ji}$$

or equivalently:

$$(6.3.4) \quad \frac{\partial X}{\partial x_{ij}^*} = \frac{1}{2}(2 - \delta_{ij})(J_{ij} + J_{ji}) \quad (\text{Browne (1979)})$$

$$(6.3.5) \quad \frac{\partial X}{\partial x_{ij}^*} = \frac{1}{1 + \delta_{ij}} (J_{ij} + J_{ji}) \quad (\text{Browne (1979)}),$$

where δ_{ij} denotes Kronecker's delta.

Combining (6.3.2) with (6.3.3) (say) it follows from (5.1.3) that:

$$(6.3.6) \quad \frac{\partial y}{\partial x_{ij}^*} = \left(\frac{\partial y}{\partial \mathbf{X}} \right)_{ij} + (1 - \delta_{ij}) \left(\frac{\partial y}{\partial \mathbf{X}} \right)_{ji}$$

Notice that this expression is *not* a typical element of $\frac{\partial y}{\partial X}$ (nor of $\frac{\partial y}{\partial \mathbf{X}}$) since X is symmetric and to conclude that $\frac{\partial y}{\partial X} = 2 \left(\frac{\partial y}{\partial \mathbf{X}} \right) - \text{Diag} \left(\frac{\partial y}{\partial \mathbf{X}} \right)$ would lead to inconsistencies.

Similarly the derivative $\frac{\partial y}{\partial \mathbf{X}}$ cannot be reconstructed from (6.3.2) for *any* patterned matrix $\mathbf{X}$. Derivatives should only be taken with respect to the mathematically independent and variable elements as emphasized in (6.2.2).

Example 6.3.2

Consider $y = |\mathbf{X}|$ where $\mathbf{X}: 2 \times 2$ is symmetric. Using the element breakdown method:

$$\begin{aligned}\frac{\partial y}{\partial x_{ij}^*} &= \text{tr} \left\{ \frac{\partial |\mathbf{X}|}{\partial \mathbf{X}} \frac{\partial \mathbf{X}}{\partial x_{ij}^*} \right\} \\ &= |\mathbf{X}| \text{tr} (\mathbf{X}^{-T} (\mathbf{J}_{ij} + (1 - \delta_{ij}) \mathbf{J}_{ji})) \\ &= |\mathbf{X}| (x^{jj} + (1 - \delta_{ij}) x^{ji})\end{aligned}$$

Thus

$$\frac{\partial y}{\partial \text{vecp}' \mathbf{X}} = |\mathbf{X}| \begin{bmatrix} x^{11} (x^{12} + x^{21}) & x^{22} \end{bmatrix}.$$

Using the vector rearrangement method:

$$\begin{aligned}\frac{\partial y}{\partial \text{vecp}' \mathbf{X}} &= \frac{\partial |\mathbf{X}|}{\partial \text{vec}' \mathbf{X}} \frac{\partial \text{vec} \mathbf{X}}{\partial \text{vecp}' \mathbf{X}} \\ &= \text{vec}' (|\mathbf{X}| \mathbf{X}^{-T}) \mathbf{K}_2^-(s)\end{aligned}$$

$$\text{where } \mathbf{K}_2^-(s) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{then:}$$

$$\frac{\partial y}{\partial \text{vecp}' \mathbf{X}} = |\mathbf{X}| \begin{bmatrix} x^{11}, (x^{12} + x^{21}), x^{22} \end{bmatrix}$$

Some remarks concerning derivatives of certain scalar valued functions of symmetric matrices

- (a) Suppose $y = y(\mathbf{X})$ denotes a function of $\mathbf{X}$ that should be maximized or minimized, like a likelihood function or a Lagrangian function. According to (6.3.6), $\frac{\partial y}{\partial x_{ij}^*} = 0$ if and only if $(\frac{\partial y}{\partial \mathbf{X}})'_{ij} = 0$. Thus if $\frac{\partial y}{\partial \mathbf{X}} = 0$ yields $\hat{\mathbf{X}}$ as a solution, and if $\hat{\mathbf{X}} = \hat{\mathbf{X}}'$, then $\hat{\mathbf{X}}$ is the maximizer or minimizer of $y(\mathbf{X})$. If $\hat{\mathbf{X}} \neq \hat{\mathbf{X}}'$ then $\hat{\hat{\mathbf{X}}} = \frac{1}{2}(\hat{\mathbf{X}} + \hat{\mathbf{X}}')$ will be the maximizer or minimizer of $y(\mathbf{X})$ since $\hat{\hat{\mathbf{X}}}'$ is also a solution of $\frac{\partial y}{\partial x_{ij}^*} = 0$ for

all $i, j = 1, \dots, p$.

- (b) A technique which is particularly suitable to differentiate a characteristic function

$\phi_A(\Theta)$ to obtain the moments of the symmetric matrix A is the following: Write the symmetric parameter matrix Θ as $\frac{1}{2}(\Phi + \Phi')$ where Φ is considered as mathematically independent and variable. Now differentiate with respect to Φ to obtain the central moments of $\frac{1}{2}(A + A') \doteq A$, when the derivatives are equated to a zero matrix. (De Waal and Nel (1973)).

7. Matrix derivatives of Kronecker products; higher derivatives and Hessian matrices

7.1 Derivatives of Kronecker products:

Neudecker (1969) introduced the concept of a derivative of a Kronecker product by differentiating Kronecker products of the following forms with respect to the i, j -th element of X :

If

$$(7.1.1) \quad Y = A \otimes BXC, \text{ then } \frac{\partial Y}{\partial x_{ij}} = A \otimes BJ_{ij}C$$

and if

$$(7.1.2) \quad Y = FXG \otimes H, \text{ then } \frac{\partial Y}{\partial x_{ij}} = FJ_{ij}G \otimes H$$

MacRae (1974) extended these results to find the Kronecker product arrangement derivative of a Kronecker product of two matrix functions of X ;

If $Y = W \otimes Z$ where $W = W(X) : s \times t$, $Z = Z(X) : p \times q$ and $X : m \times n$, then;

$$(7.1.3) \quad \frac{\partial Y}{\partial X} = W \otimes \frac{\partial Z}{\partial X} + (I_{(p,s)} \otimes I_m) (Z \otimes \frac{\partial W}{\partial X}) (I_{(t,q)} \otimes I_n).$$

McDonald (1976) defined the matrix derivative of a simple Kronecker product and then extended this vector rearrangement derivative to other Kronecker products e.g.

If $Y = X \otimes I_r$ where $X : m \times n$ is mathematically independent and variable:

$$(7.1.4) \quad \frac{\partial \text{vec} Y}{\partial \text{vec} X} = (I_r \otimes I_{(n,m)} \otimes I_r) (I_{mn} \otimes \text{vec} I_r).$$

If $Y = X \otimes B$ where $B : s \times r$:

$$(7.1.5) \quad \frac{\partial \text{vec} Y}{\partial \text{vec} X} = (I_{nr} \otimes I_m \otimes B) \frac{\partial \text{vec}(X \otimes I_r)}{\partial \text{vec} X}.$$

A general Kronecker product rule given by McDonald (1976) is:

If $Z = M(Y \otimes B)N$ where $M: t \times u$, $N: v \times s$, $Y = Y(X): u \times v$, $B: r \times s$ and if $X: m \times n$ is mathematically independent and variable:

$$(7.1.6) \quad \frac{\partial \text{vec} Z}{\partial \text{vec}' X} = N' \otimes M (I_{vs} \otimes I_{u \otimes B}) \frac{\partial \text{vec}(Y \otimes I_s)}{\partial \text{vec}' Y} \frac{\partial \text{vec} Y}{\partial \text{vec}' X}.$$

Bentler and Lee (1978) give a vector rearrangement derivative which is equivalent to (7.1.3):

$$(7.1.7) \quad \frac{\partial \text{vec}(W \otimes Z)}{\partial \text{vec}' X} = (I_t \otimes I_{(s, t)} \otimes I_p)(I_{st} \otimes (\text{vec} Z)') \frac{\partial \text{vec} W}{\partial \text{vec}' X} \\ + (\text{vec} W')' \frac{\partial \text{vec} Z}{\partial \text{vec}' X} \otimes I_{pq}$$

7.2 Higher derivatives and Hessian matrices

If $Y: p \times q = Y(x, z)$ is a matrix valued function of the scalars x and z , the second order derivative of Y with respect to x and z is defined as:

$$(7.2.1) \quad \frac{\partial^2 Y}{\partial x \partial z} = \frac{\partial}{\partial x} \left(\frac{\partial Y}{\partial z} \right) = \frac{\partial}{\partial z} \left(\frac{\partial Y}{\partial x} \right) = \frac{\partial^2 Y}{\partial z \partial x}$$

with typical element $\left(\frac{\partial^2 Y}{\partial x \partial z} \right)_{ij} = \frac{\partial^2 y_{ij}}{\partial x \partial z}.$

The chain rule (5.1.2) can be used to obtain successive derivatives when using the element breakdown method. The element breakdown method is particularly suitable to obtain higher derivatives as is illustrated in the following examples:

Example 7.2.1

If $y = \text{tr}(AX^{-1})$ then according to (5.1.2):

$$\frac{\partial y}{\partial x_{ij}} = \text{tr} \left\{ \frac{\partial \text{tr} AX^{-1}}{\partial X^{-1}} \frac{\partial X^{-1}}{\partial x_{ij}} \right\} = \text{tr} \{ A' (-X^{-T} J_{ji} X^{-T}) \}$$

$$= \text{tr} \{ -X^{-T} A' X^{-T} J_{ji} \} = -\text{tr} \{ J_{ji} X^{-1} A X^{-1} \}.$$

By using (5.1.2) again:

$$\frac{\partial^2 y}{\partial x_{ij} \partial x_{gh}} = \text{tr} \left\{ \frac{\partial \text{tr} (-J_{ji} X^{-1} A X^{-1})}{\partial (X^{-1} A X^{-1})} \frac{\partial (X^{-1} A' X^{-1})}{\partial x_{gh}} \right\}$$

$$\begin{aligned}
\frac{\partial^2 y}{\partial x_{ij} \partial x_{gh}} &= \text{tr} \{ -J_{ji} (-X^{-T} J_{hg} X^{-T} A' X^{-T} - X^{-T} A' X^{-T} J_{hg} X^{-T}) \} \\
&= \text{tr} \{ J_{ji} X^{-T} J_{hg} X^{-T} A' X^{-T} + J_{ji} X^{-T} A' X^{-T} J_{hg} X^{-T} \} \\
&= (X^{-T})_{jh} (X^{-T} A' X^{-T})_{gi} + (X^{-T} A' X^{-T})_{jh} (X^{-T})_{gi} \quad (\text{from 5.1.4})
\end{aligned}$$

from which $\frac{\partial y}{\partial \text{vec} X \partial \text{vec}' X}$ can be constructed. (See (7.2.4), (7.2.5) and Example 7.2.2).

If $y = y(\underline{x}, \underline{z})$ is a scalar valued function of the vector variables $\underline{x}: p \times 1$ and $\underline{z}: q \times 1$, then $\frac{\partial^2 y}{\partial \underline{x} \partial \underline{z}'}$ is the $p \times q$ matrix defined by:

$$(7.2.2) \quad \frac{\partial^2 y}{\partial \underline{x} \partial \underline{z}'} = \frac{\partial}{\partial \underline{z}'} \left(\frac{\partial y}{\partial \underline{x}} \right)$$

with a typical element $\left(\frac{\partial^2 y}{\partial \underline{x} \partial \underline{z}'} \right)_{ij} = \frac{\partial^2 y}{\partial x_i \partial z_j}$ (Browne (1979)).

If $y = y(X, Z)$ is a scalar valued function of the mathematically independent matrix variables $X: m \times n$ and $Z: p \times q$, then $\frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z}$ is an $mn \times pq$ matrix defined by:

$$(7.2.3) \quad \frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z} = \frac{\partial}{\partial \text{vec}' Z} \left(\frac{\partial y}{\partial \text{vec} X} \right)$$

with a typical element $\left(\frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z} \right)_{ij, gh} = \frac{\partial^2 y}{\partial x_{ij} \partial z_{gh}}$. (Neudecker (1969).

If X, Z, A and $B: p \times p$ then

$\frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z}$ can be constructed from $\frac{\partial^2 y}{\partial x_{ij} \partial z_{gh}}$ by using the following recognition rules:

$$(7.2.4) \quad \text{If } \frac{\partial^2 y}{\partial x_{ij} \partial z_{gh}} = a_{jh} b_{ig} \quad \text{then} \quad \frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z} = A \otimes B. \quad (\text{Browne (1974)}).$$

$$(7.2.5) \quad \text{If } \frac{\partial^2 y}{\partial x_{ij} \partial z_{gh}} = a_{ih} b_{jg} \quad \text{then} \quad \frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' Z} = I_{(p, p)} (A \otimes B).$$

Example 7.2.2

From $\frac{\partial^2 y}{\partial x_{ij} \partial x_{gh}}$ in example 7.2.1, it follows by using (7.2.5) that:

$$\frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' X} = I_{(p,p)} [(X^{-T} \otimes X^{-1} A X^{-1}) + (X^{-T} A' X^{-T} \otimes X^{-1})] .$$

It should be noted however that the second derivatives in the form $\frac{\partial^2 y}{\partial x_{ij} \partial x_{gh}}$ are far more convenient for computer application than the alternative given by $\frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' X}$.

If $\underline{x}$:px1 is mathematically independent and variable, and $y = y(\underline{x})$ a scalar valued function of $\underline{x}$, then the pxp matrix:

$$(7.2.6) \quad H = H(\underline{x}) = \frac{\partial^2 y}{\partial \underline{x} \partial \underline{x}'}$$

is known as the Hessian of y and also denoted as $\nabla_{\underline{x}}^2 y$.

If $y = y(X)$ is a scalar valued function of X :mxn the Hessian $H(X)$ is given by:

$$(7.2.7) \quad \nabla_{\text{vec} X}^2 y = \frac{\partial^2 y}{\partial \text{vec} X \partial \text{vec}' X} \quad (\text{Neudecker (1969), (1975)}).$$

A good convention for finding Hessians are:

$$(7.2.8) \quad \text{If } d^2\phi = (d\underline{x})' A (d\underline{y}), \text{ then } \frac{\partial^2 \phi}{\partial \underline{x} \partial \underline{y}'} = A.$$

$$(7.2.9) \quad \text{If } d^2\phi = (d\underline{x})' A (d\underline{x}), \text{ then } \frac{\partial^2 \phi}{\partial \underline{x} \partial \underline{x}'} = \frac{1}{2}(A + A'). \quad (\text{Neudecker 1969}).$$

There are basically two distinct expressions for $d^2\phi$ from which Hessians

$$\frac{\partial^2 \phi}{\partial \text{vec} X \partial \text{vec}' Y}$$

can be derived, a.g.:

If X :mxn and Y :pxq are mathematically independent and variable then the recognition rules are:

$$(7.2.10) \quad d^2\phi = \text{tr}(A(dX) B(dY)C) \text{ implies } \frac{\partial^2 \phi}{\partial \text{vec} X \partial \text{vec}' Y} = I_{(m,n)} (A' C' \otimes B)$$

$$(7.2.11) \quad d^2\phi = \text{tr}(A(dX) B(dY)'C) \text{ implies } \frac{\partial^2\phi}{\partial \text{vec} X \partial \text{vec}' Y} = B \otimes A' C'.$$

(A list of similar Hessians are given in Tracy and Singh (1973).)

The proof of (7.2.10) is given, since it is informative:

$$\begin{aligned} d^2\phi &= \text{tr}(A(dX) B(dY)C) = \text{tr}(dX) B(dY)CA \\ &= (\text{dvec} X')' (A' C' \otimes B) (\text{dvec} Y), \text{ from (4.2.6)} \\ &= (\text{dvec} X')' I_{(m, n)} (A' C' \otimes B) (\text{dvec} Y), \text{ from (4.2.2)}. \end{aligned}$$

Thus

$$\frac{\partial^2\phi}{\partial \text{vec} X \partial \text{vec}' Y} = I_{(m, n)} (A' C' \otimes B) = (A' C' \otimes B)_{(n)}, \quad \text{from (4.1.3). Now using}$$

(7.2.9), it follows from (7.2.10) and (7.2.11) respectively:

$$(7.2.12) \quad d^2\phi = \text{tr}(A(dX) B(dX)C) \text{ implies}$$

$$\frac{\partial^2\phi}{\partial \text{vec} X \partial \text{vec}' X} = \frac{1}{2} (I_{(m, n)} (A' C' \otimes B) + I_{(n, m)} (CA \otimes B')).$$

$$(7.2.13) \quad d^2\phi = \text{tr}(A(dx) B(dx)'C) \text{ implies } \frac{\partial^2\phi}{\partial \text{vec} X \partial \text{vec}' X} = \frac{1}{2} (B \otimes A' C' + B' \otimes CA)$$

(see also Neudecker (1969)).

Following McDonald (1976), the second derivative of a matrix valued function $Y = Y(X, Z)$:sxt of the matrices X :mxn and Z :pxq can be defined as the stmn x pq matrix

$$\frac{\partial}{\partial \text{vec}' Z} (\text{vec}(\frac{\partial \text{vec} Y}{\partial \text{vec}' X})).$$

If $s = t = 1$ this derivative reduces to (7.2.3)

McDonald (1976) also derived formulae to find $\frac{\delta^2\phi}{\delta X \delta Y}$, where ϕ is a scalar valued function of X :mxn and Y :pxq. Here $\frac{\delta Y}{\delta X}$ will denote $\frac{\partial \text{vec}' Y'}{\partial \text{vec} X'}$ and $\frac{\delta^2\phi}{\delta X \delta Y} = \frac{\partial}{\partial \text{vec} X'} (\frac{\partial \phi}{\partial \text{vec}' Y'})$.

If $\phi = \Phi(W, Z)$ where $W = W(X, Y)$: txu and $Z = Z(X, Y)$: rxn then:

$$(7.2.14) \quad \frac{\delta^2 \phi}{\delta X \delta Y} = \frac{\delta W}{\delta X} \frac{\delta^2 \phi}{\delta W^2} \left(\frac{\delta W}{\delta Y} \right)' + \frac{\delta Z}{\delta X} \frac{\delta^2 \phi}{\delta Z^2} \left(\frac{\delta Z}{\delta Y} \right)' + \frac{\delta W}{\delta X} \frac{\delta^2 \phi}{\delta W \delta Z} \left(\frac{\delta Z}{\delta Y} \right)' \\ + \frac{\delta Z}{\delta X} \frac{\delta^2 \phi}{\delta Z \delta W} \left(\frac{\delta W}{\delta Y} \right)' + \frac{\delta^2 W}{\delta X \delta Y} (I_{pq} \otimes \frac{\delta \phi}{\delta W}) + \frac{\delta^2 Z}{\delta X \delta Y} (I_{pq} \otimes \frac{\delta \phi}{\delta Z}).$$

Similar results are also given for the cases where:

$$\phi = \phi(W, Z), \quad W = W(X) \quad \text{and} \quad Z = Z(Y),$$

$$\phi = \phi(W), \quad W = W(X, Y) \quad \text{and}$$

$$\phi = \phi(W), \quad W = F_1(X) + F_2(Y). \quad \text{Identities to simplify expres-}$$

sions like $\frac{\delta^2 Z}{\delta X \delta Y} (I_{pq} \otimes \frac{\delta \phi}{\delta Z})$ appearing in these derivatives are also given by McDonald.

8. Derivatives of Partitioned Matrices

8.1 The double Kronecker product:

Rao and Mitra (1971), Tracy and Singh (1972 (b)) and McDonald and Swaminathan (1973) defined a new matrix product, closely related to the Kronecker matrix product, which will be called the double Kronecker matrix product and be denoted by $A \odot B$ according to Rao and Mitra (1971).

The more general definition, given by Tracy and Singh (1972) is the following:

Suppose $A: m \times n$ is partitioned into vr submatrices:

$A_{ij}: m_i \times n_j$ as $A = [A_{ij}]$, where $\sum_{i=1}^v m_i = m$ and $\sum_{j=1}^r n_j = n$. Also suppose $B: p \times q$ is partitioned into gk submatrices $B_{st}: p_s \times q_t$ as $B = [B_{st}]$, where $\sum_{s=1}^g p_s = p$ and $\sum_{t=1}^k q_t = q$.

The submatrix in the s, t -position of the product $A \odot B$ is then the $m p_s \times n q_t$ matrix:

$$(A \odot B)_{s, t} = [A_{ij} \otimes B_{st}] : m p_s \times n q_t \\ = \begin{bmatrix} A_{11} \otimes B_{st} & \dots & A_{1r} \otimes B_{st} \\ \dots & \dots & \dots \\ A_{p1} \otimes B_{st} & \dots & A_{pr} \otimes B_{st} \end{bmatrix} \begin{matrix} m_1 p_s \\ \vdots \\ m_v p_s \end{matrix} \\ n_1 q_t \quad \dots \quad n_r q_t$$

such that the double Kronecker matrix product is:

$$(8.1.1) \quad A \odot B: mp \times nq = \begin{bmatrix} (A \odot B)_{11} & \dots & (A \odot B)_{1k} \\ \dots & \dots & \dots \\ (A \odot B)_{g1} & \dots & (A \odot B)_{gk} \end{bmatrix} \begin{matrix} mp_1 \\ \vdots \\ mp_g \\ nq_1 & \dots & nq_k \end{matrix}$$

The laws governing this product are similar to the laws governing the Kronecker product e.g.

$$(8.1.2) \quad (A \odot B)' = A' \odot B'; \quad A \odot (B \odot C) = (A \odot B) \odot C \\ A \odot (B+C) = A \odot B + A \odot C; (A \odot B)(C \odot D) = AC \odot BD.$$

8.2 The vector rearrangement method for partitioned matrices:

It should be noted that the methods used in this section are similar to those described in section 5.3 with the exception that the vec operator is now defined to take submatrices into account too. The role played by the Kronecker product is now taken over by the double Kronecker product. Although Tracy and Singh (1972 (b)) and McDonald and Swaminathan (1973) developed this method for rowpacked vectors $\underline{Y}_R$ we shall in line with the previous sections give the results in the column packed vector rearrangement form, now denoted as VECY or $\underline{Y}_C$.

Definition 8.2.1

If $Y: pq$ is partitioned into submatrices as:

$$Y = \begin{bmatrix} Y_{11} & Y_{12} & \dots & Y_{1n} \\ \dots & \dots & \dots & \dots \\ Y_{m1} & Y_{m2} & \dots & Y_{mn} \end{bmatrix} \begin{matrix} p_1 \\ \dots \\ p_m \\ q_1 & q_2 & \dots & q_n \end{matrix}$$

then by $\text{VECY} = \underline{Y}_C$, we will denote the column vector:

$\text{VECY}: pq \times 1 = [\text{vec}'Y_{11}, \text{vec}'Y_{21}, \dots, \text{vec}'Y_{m1}, \dots, \text{vec}'Y_{1n}, \dots, \text{vec}'Y_{mn}]'$, and by

$\text{VECY}' = \underline{Y}_R$, the column vector:

$\text{VECY}': pq \times 1 = [\text{vec}'Y'_{11}, \text{vec}'Y'_{12}, \dots, \text{vec}'Y'_{1n}, \dots, \text{vec}'Y'_{m1}, \dots, \text{vec}'Y'_{mn}]'$.

Tracy and Singh (1972 (b)) defined $p \times p \times q$ matrices (denoted $\{m\} I$ and $I_{\{n\}}$), which we will denote as $I_{\{n, m\}}$ and $I_{\{m, n\}}$ and which respectively serve the same purpose as the permuted identity matrices $I_{(n, m)}$ and $I_{(m, n)}$ (see sections 4.1 and 4.2) such that:

$$(8.2.1) \quad \text{VECY} = I_{\{m, n\}} \text{VECY}' \text{ and } \text{VECY}' = I_{\{n, m\}} \text{VECY}.$$

These matrices $I_{\{n, m\}}$ and $I_{\{m, n\}}$ have analogous properties to $I_{(n, m)}$ and $I_{(m, n)}$ with the Kronecker product replaced by the double Kronecker product and where m and n refer to the number of row and column submatrices of Y e.g. If $Y = AXB$, then:

$$(8.2.2) \quad \text{VECY} = (B' \odot A) \text{VECX}, \text{ and if } Y = CX'D \text{ then:}$$

$$(8.2.3) \quad \text{VECY} = (D' \odot C) I_{\{n, m\}} \text{VECX}, \text{ analogous to (4.2.3) and (4.2.4).}$$

Similarly to (5.3.2), if:

$$(8.2.4) \quad \text{VECY} = H \text{VECX} \text{ then } \frac{\partial \text{VECY}}{\partial \text{VEC}'X} = H.$$

Similar results to (5.3.4) to (5.3.7) can now be proved for $\frac{\partial \text{VECY}}{\partial \text{VEC}'X}$, for example:

If $Y = AXB + CX'D$, then $d\text{VECY} = ((B' \odot A) + (D' \odot C) I_{\{n, m\}}) d\text{VECX}$ and consequently: $\frac{\partial \text{VECY}}{\partial \text{VEC}'X} = B' \odot A + (D' \odot C) I_{\{n, m\}}$.

McDonald and Swaminathan (1973) extended the chain rule (5.3.10) to:

$$(8.2.5) \quad \frac{\partial \text{VECY}}{\partial \text{VEC}'X} = \frac{\partial \text{VECY}}{\partial \text{VEC}'Y} \frac{\partial \text{VECY}}{\partial \text{VEC}'X}$$

where X is mathematically independent too in the sense that all submatrices of X are functionally independent and contain no constant elements. If X does contain constant submatrices i.e. if X is block-patterned then an analogous theory to section 6 can be considered with block-patterned transition and pattern matrices.

Swaminathan (1976) generalized McDonald's result (7.2.9) for second derivatives to second derivatives for partitioned matrices e.g.:

If $\phi = \phi(W_1, \dots, W_k)$ is a scalar valued function of the matrices $W_1, \dots, W_k$, where $W_i = W_i(X_1, \dots, X_k)$, $i=1, \dots, k$ are matrix valued functions of the matrices $X_1, \dots, X_k$, then:

$$(8.2.7) \quad \frac{\partial^2 \phi}{\partial \text{VEC} X_i' \partial \text{VEC}' X_j'} = \sum_{m=1}^k \frac{\partial^2 \text{VEC}' W_m'}{\partial \text{VEC} X_i' \partial \text{VEC}' X_j'} \left(\text{IO} \frac{\partial \phi}{\text{VEC} W_m'} \right) \\ + \sum_{m=1}^k \sum_{\ell=1}^k \frac{\partial \text{VEC}' W_\ell'}{\partial \text{VEC} X_i'} \frac{\partial^2 \phi}{\partial \text{VEC} W_\ell' \partial \text{VEC}' W_m'} \frac{\partial \text{VEC}' W_m'}{\partial \text{VEC} X_j'}$$

9. The evaluation of Jacobians

A recent development in Multivariate Analysis is to determine the Jacobians of matrix bijections by using matrix differentiation. Many of the Jacobians which will be considered here have been proved by Hsu (1953), whose notes on Jacobians were published by Deemer and Olkin (1951). Olkin (1953), Olkin and Roy (1954), Roy (1952) (1957), Olkin and Sampson (1972), Jack (1964-1965) extended these results to Jacobian's of nonlinear and orthogonal transformations. However these authors did not use matrix derivatives to derive these Jacobians.

Aitken (1949) mentioned the idea of using the vec operator in connection with Jacobians but Neudecker (1969) used matrix differentiation in the derivation of Jacobians by deriving the Jacobian of the bijection $Y = AXB$, where all matrices are $n \times n$.

He defined: $J(Y \rightarrow X) = \left| \frac{\partial \text{vec}' Y}{\partial \text{vec} X} \right|$. Tracy and Singh (1972 (a)) extended this idea to

symmetric matrices e.g. $J(Y \rightarrow X) = \left| \frac{\partial \text{vecp} Y}{\partial \text{vecp} X} \right|$, but their proofs lacked mathematical detail. In analogy to their definition we define the Jacobian of a bijection between two patterned matrices as follows:

Definition 9.1

Suppose $Y: p \times q$ is a patterned matrix with p^* mathematically independent elements and with pattern (1). Suppose $X: m \times n$ has pattern (2) but also p^* mathematically independent elements and that a bijection exists between the mathematical independent elements of these matrices, then the Jacobian of this bijection $Y \rightarrow X$ is given by:

$$J(Y \rightarrow X) = \left| \frac{\partial \text{vecp}_1 Y}{\partial \text{vecp}_2 X} \right|.$$

If Y and X are mathematically independent matrices, both of order $p \times q$ then

$$J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right|. \text{ If } q=1, \text{ i.e. } Y \text{ and } X \text{ are vectors then}$$

$$J(\underline{y} \rightarrow \underline{x}) = \left| \frac{\partial \underline{y}}{\partial \underline{x}^T} \right|$$

The rules $J(Y \rightarrow X) = J(Y \rightarrow X) J(Z \rightarrow X)$ and $J(Y \rightarrow X) = (J(X \rightarrow Y))^{-1}$ follow from (3.2.3), (3.2.6) and their counterparts in section 5.3, by taking determinants.

9.1 Useful determinants for the evaluation of Jacobians

The following results are useful for determining Jacobians.

$$(9.1.1) \quad \text{If } A: p \times p \text{ and } B: q \times q \text{ then } |A \otimes B| = |A|^q |B|^p.$$

Let $A: p \times p$ and $B: p \times p$ be symmetric matrices with the property $AB = BA$, so that they are simultaneously diagonalizable.

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ and $\delta_1 \geq \delta_2 \geq \dots \geq \delta_p$ denote the eigenvalues of A and B respectively, then if K_p is the transition matrix for any patterned matrix $X: p \times p$.

$$(9.1.2) \quad |K_p^-(A \otimes B)K_p| = |K_p^-(\Lambda \otimes \Delta)K_p|$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ and $\Delta = \text{diag}(\delta_1, \dots, \delta_p)$. The proof is similar to the proof given for the symmetric case. (Nel (1978)). If $K_p(s)$ is the symmetric transition matrix, then using (9.1.2):

$$(9.1.3) \quad |K_p^-(s)(A \otimes B)K_p(s)| = 2^{-\frac{1}{2}p(p+1)} \prod_{i \leq j} (\lambda_i \delta_j + \lambda_j \delta_i)$$

(Nel (1978), Neudecker (1978)).

If $B = A$, then

$$(9.1.4) \quad |K_p^-(s)(A \otimes A)K_p(s)| = \prod_{i \leq j} \lambda_i \lambda_j = |A|^{p+1}.$$

Henderson and Searle (1979 (b)) also proved (9.1.4) for the case when A is not a simple matrix by using the singular value decomposition of A .

$$\text{If } B = I_p: |K_p^-(s)(A \otimes I_p)K_p(s)| = 2^{-\frac{1}{2}p(p+1)} \prod_{i \leq j} (\lambda_i + \lambda_j)$$

and since $K_p^-(s)(A \otimes I_p)K_p(s) = K_p^-(s)(I_p \otimes A)K_p(s)$:

$$(9.1.5) \quad |K_p^-(s)(A \otimes I_p + I_p \otimes A)K_p(s)| = \prod_{i \leq j} (\lambda_i + \lambda_j).$$

Similarly it follows that if K_p denotes $K_p(ss)$ or $K_p(c)$:

$$(9.1.6) \quad |K_p^-(A \otimes B)K_p| = 2^{\frac{1}{2}P(P-1)} \prod_{i < j} (\lambda_i \delta_j + \lambda_j \delta_i),$$

$$(9.1.7) \quad |K_p^-(A \otimes A)K_p| = \prod_{i < j} \lambda_i \lambda_j = |A|^{P-1},$$

$$(9.1.8) \quad |K_p^-(A \otimes I_p + I_p \otimes A)K_p| = \prod_{i < j} (\lambda_i + \lambda_j).$$

If A and B are lower triangular, and if $K_p(\ell)$ is the lower triangular transition matrix:

$$(9.1.9) \quad |K_p'(\ell)(A \otimes B)K_p(\ell)| = |K_p'(\ell)(A \otimes B')K_p(\ell)| = \prod_{i=1}^P a_{ii}^{p-i+1} b_{ii}^i$$

$$\begin{aligned} (9.1.10) \quad |K_p'(\ell)(I_p^2 + I_{(p,p)})(A \otimes B)K_p(\ell)| &= |2K_p'(\ell)M_p(s)(A \otimes B)K_p(\ell)| \\ &= |2K_p'(s)(A \otimes B)K_p(\ell)| \\ &= 2^P \prod_{i=1}^P a_{ii}^{p-i+1} b_{ii}^i, \end{aligned}$$

since $M_p(s) = \frac{1}{2}(I_p^2 + I_{(p,p)})$ and $M_p(s)K_p(1) = K_p(s)$.

Similarly:

$$(9.1.11) \quad |2K_p'(s)(A' \otimes B)K_p(\ell)| = 2^P \prod_{i=1}^P a_{ii}^{p-i+1} b_{ii}^i.$$

The results (9.1.9) to (9.1.11) were proved by Neudecker (1978). A simple expression for $|2K_p'(s)(A \otimes B'(K_p(\ell)))|$ does not exist.

From (9.1.4) and (9.1.7) we conjecture that for any transition matrix K_p for which (6.1.6) is valid for the corresponding pattern matrix M_p :

$$|K_p^-(A \otimes A)K_p| = |A|^{\frac{2}{P}P^*}$$

where p^* is the number of mathematically independent and variable elements of the patterned matrix X .

9.2 Jacobians of certain matrix transformations by matrix differentiation

In this section the vector — and vector rearrangement derivatives are used to derive the Jacobians of matrix bijections.

If $\underline{y}:p \times 1 = A \underline{x}$, where $\underline{x}:p \times 1$ then:

$$(9.2.1) \quad J(\underline{y} \rightarrow \underline{x}) = \left| \frac{\partial \underline{y}}{\partial \underline{x}'} \right| = |A|.$$

If $Y = AXB$ where $Y, X: p \times q$ then:

$$(9.2.2) \quad J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |B' \otimes A| = |B|^p |A|^q.$$

If $A = I_p$, then $Y = XB$ and $J(Y \rightarrow X) = |B|^p$.

If $B = I_q$, then $Y = AX$ and $J(Y \rightarrow X) = |A|^q$.

If $A = cI_p, B = I_q$ then $Y = cX$ and $J(Y \rightarrow X) = c^{pq}$.

(Tracy and Singh (1972) (a)).

If $Y = AX'B$, where $Y, X: p \times p$ then:

$$(9.2.3) \quad J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |(B' \otimes A)I_{(p, p)}| = (-1)^{\frac{1}{2}p(p-1)} |A|^p |B|^p,$$

since $|I_{(p,p)}| = (-1)^{\frac{1}{2}p(p-1)}$, (Magnus and Neudecker (1979)).

If $Y = X^{-1}$ where Y, X are mathematically independent and variable:

$$(9.2.4) \quad J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |X^{-T} \otimes X^{-1}| = |X|^{-2p}$$

If $Y = X^n: p \times p$ where X is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_p$,

$$\begin{aligned}
 (9.2.5) \quad J(Y \rightarrow X) &= \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = \left| \sum_{s=1}^n (X')^{n-s} \otimes X^{s-1} \right| \quad (\text{from (5.3.7)}) \\
 &= \left| \sum_{s=1}^n (\Lambda^{n-s} \otimes \Lambda^{s-1}) \right|, \text{ where } H^{-1}XH = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p) \\
 &= \prod_{i=1}^p \prod_{j=1}^p \sum_{s=1}^n \lambda_i^{n-s} \lambda_j^{s-1}, (\text{Henderson and Searle (1979 (b))})
 \end{aligned}$$

since if $a_1, \dots, a_p$ are the eigenvalues of A and $b_1, \dots, b_n$ the eigenvalues of B , then $a_i b_j$ ($i=1, \dots, p; j=1, \dots, n$) are the eigenvalues of $A \otimes B$ and $|A \otimes B| = \prod_{i=1}^p \prod_{j=1}^n a_i b_j$.

If $Y, X: p \times p$ are *symmetric* and $Y = AXA'$, then:

$$(9.2.6) \quad J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |K_p^-(s)(A \otimes A)K_p(s)| = |A|^{p+1}, \text{ from (9.1.4).}$$

If $A = A': p \times p$ and $Y = XAX'$, then:

$$(9.2.7) \quad J(Y \rightarrow X) = \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |K_p^-(s)(I \otimes AX + AX \otimes I)K_p(s)| = \prod_{i \leq j} (v_i + v_j),$$

from (9.1.5). (Tracy and Singh (1972 (a)), Neudecker (1978)).

If $A, B: p \times p$ and $Y = AXA' + BXB'$, then:

$$\begin{aligned}
 (9.2.8) \quad J(Y \rightarrow X) &= \left| \frac{\partial \text{vec} Y}{\partial \text{vec}' X} \right| = |K_p^-(s)(A \otimes A + B \otimes B)K_p(s)| \\
 &= \prod_{i \leq j} (\lambda_i \lambda_j + \delta_i \delta_j). \quad (\text{Tracy and Singh (1972 (a)), Neudecker (1978)}).
 \end{aligned}$$

If $Y = X^n: p \times p$ where X is symmetric with eigenvalues $\lambda_1, \dots, \lambda_p$,

$$\begin{aligned}
 (9.2.9) \quad J(Y \rightarrow X) &= |K_p^-(s) \sum_{s=1}^n (\Lambda^{n-s} \otimes \Lambda^{s-1}) K_p(s)| \\
 &= \prod_{i=1}^p \prod_{j=i}^p \sum_{s=1}^n \lambda_i^{n-s} \lambda_j^{s-1}
 \end{aligned}$$

(Henderson and Searle (1979(b))).

If $Y = X^{-1}$ where $X = X'$ and nonsingular, then:

$$\begin{aligned}
 (9.2.10) \quad J(Y \rightarrow X) &= \left| \frac{\partial \text{vecp} Y}{\partial \text{vecp} X} \right| = \left| \frac{\partial \text{vecp} Y}{\partial \text{vec} Y} \frac{\partial \text{vec} Y}{\partial \text{vec} X} \frac{\partial \text{vec} X}{\partial \text{vecp} X} \right| \\
 &= |K_p'(s)(-X^{-T} \otimes X^{-1})K_p^{-1}(s)|, \text{ from (6.2.1) and (5.3.7),} \\
 &= |-K_p^{-1}(s)(X^{-1} \otimes X^{-1})K_p(s)| = (-1)^{\frac{1}{2}p(p+1)} |X|^{-(p+1)}, \\
 &\text{from (9.1.4). (Tracy and Singh (1972 (a)), Nel (1976),} \\
 &\text{Neudecker (1978)).}
 \end{aligned}$$

If $Y = |X|X^{-1}$, $X = X'$ and nonsingular, then:

$$\begin{aligned}
 (9.2.11) \quad J(Y \rightarrow X) &= |K_p'(s)|X|(\text{vec}(X^{-1}) \text{vec}'(X^{-1}) - X^{-T} \otimes X^{-1})K_p^{-1}(s)| \\
 &= |X|^{\frac{1}{2}p(p+1)} |K_p^{-1}(s)(\{\text{vec}(X^{-1}) \text{vec}'(X^{-1})(X \otimes X - I_p 2)\} \\
 &\quad (X^{-1} \otimes X^{-1}))K_p(s)| \\
 &= |X|^{\frac{1}{2}p(p+1)} |K_p^{-1}(s)(\text{vec}(X^{-1}) \text{vec}'(X) - I_p 2)K_p(s)| \\
 &\quad |K_p^{-1}(s)(X^{-1} \otimes X^{-1})K_p(s)|, \text{ since} \\
 &\quad M_p(s)(A \otimes A) = (A \otimes A)M_p(s) \text{ and } K_p(s) = M_p(s)K_p(s). \\
 &= |X|^{\frac{1}{2}p(p+1)} |X|^{-(p+1)} |K_p^{-1}(s) \text{vec}(X^{-1}) \text{vec}'(X)K_p(s) - I_{\frac{1}{2}p(p+1)}| \\
 &= |X|^{\frac{1}{2}(p+1)(p-2)} (\text{vec}'X K_p(s)K_p^{-1}(s) \text{vec}X^{-1} - 1), \\
 &\quad \text{since } |AB - I| = |BA - I|. \\
 &= |X|^{\frac{1}{2}(p+1)(p-2)} (\text{tr } XX^{-1} - 1), \text{ from (6.1.3) and (4.2.5),} \\
 &= (p-1)|X|^{\frac{1}{2}(p+1)(p-2)}.
 \end{aligned}$$

(The proof is due to Neudecker (1978) and is given here since it illustrates techniques used).

If $Y, X: p \times p$ are skew-symmetric or has a correlation pattern then:

$$\begin{aligned}
 (9.2.12) \quad \text{If } Y = AXA', J(Y \rightarrow X) &= |K_p'(A \otimes A)K_p^{-1}| = |K_p^{-1}(A' \otimes A')K_p| \\
 &= |A|^{p-1}, \text{ from (9.1.7).}
 \end{aligned}$$

$$(9.2.13) \quad \text{If } Y = X^{-1}, J(Y \rightarrow X) = |-K_p'(X^{-T} \otimes X^{-1})K_p^{-1}| = |-K_p^{-1}(X^{-1} \otimes X^{-T})K_p|$$

$$\begin{aligned}
&= (-1)^{\frac{1}{2}p(p-1)} |X|^{-(p-1)}, \text{ if } X \text{ has a correlation} \\
&\quad \text{pattern.} \\
&= (-1)^{p(p-1)} |X|^{-(p-1)} = |X|^{-(p-1)}, \text{ if } X \text{ is} \\
&\quad \text{skew-symmetric and } p \text{ is even,}
\end{aligned}$$

and where K_p denotes $K_p(ss)$ or $K_p(c)$.

In general we conjecture that if $Y = X^{-1}$ then

$$J(Y \rightarrow X) = |X|^{-\frac{2}{p}p^*},$$

where p^* is the number of mathematically independent and variable elements of X .

If Y, X, A and $B: p \times p$ are *lower triangular matrices* such that $Y = BXA$ then:

$$\begin{aligned}
(9.2.15) \quad J(Y \rightarrow X) &= |K'_p(\ell)(A' \otimes B)K_p^{-'}(\ell)| = |K_p^{-}(\ell)(A \otimes B')K_p(\ell)| \\
&= \prod_{i=1}^p a_{ii}^{p-i+1} b_{ii}^i, \text{ from (9.1.11) (Neudecker (1978)).}
\end{aligned}$$

$$\text{If } A = I_p, Y = BX \text{ and } J(Y \rightarrow X) = \prod_{i=1}^p b_{ii}^i.$$

$$\text{If } B = I_p, Y = XA \text{ and } J(Y \rightarrow X) = \prod_{i=1}^p a_{ii}^{p-i+1}.$$

Let $Y: p \times p$ be *symmetric* and A, B and $X: p \times p$ be *lower triangular*, and denote $\text{vecp}Y$ as $\text{vecps}Y$ and $\text{vecp}X$ as $\text{vecpl}X$.

$$(9.2.16) \quad \text{If } Y = XX':$$

$$\begin{aligned}
J(Y \rightarrow X) &= \left| \frac{\partial \text{vecps}Y}{\partial \text{vecpl}X} \right| \text{ (see definition 9.1)} \\
&= \left| \frac{\partial \text{vecps}Y}{\partial \text{vec}^t Y} \frac{\partial \text{vec} Y}{\partial \text{vec}^t X} \frac{\partial \text{vec} X}{\partial \text{vecpl}X} \right| \\
&= |K'_p(s)(X \otimes I_p + (I_p \otimes X)I_{(p,p)})K_p^{-'}(\ell)|, \text{ from (5.3.5)} \\
&= |K'_p(s)(I_p^2 + I_{(p,p)})(X' \otimes I_p)K_p(\ell)|, K_p^{-}(\ell) = K'_p(\ell) \\
&= |2K'_p(s)(X' \otimes I_p)K_p(\ell)|, \text{ since } M_p(s) = \frac{1}{2}(I_p^2 + I_{(p,p)}) \\
&\quad \text{and } M_p(s)K_p(s) = K_p(s).
\end{aligned}$$

$$= 2^p \prod_{i=1}^p x_{ii}^{p-i+1}, \text{ from (9.1.11). (Neudecker (1978)).}$$

$$(9.2.17) \quad \text{If } Y = X'X:$$

$$\begin{aligned} J(Y \rightarrow X) &= |K'_p(s)((X' \otimes I)(I_{(p,p)} + (I \otimes X'))K_p^{-1}(\ell)| \\ &= |K'_p(s)(I_{(p,p)} + I_p 2)(I \otimes X')K_p(\ell)| \\ &= |2K'_p(s)(I \otimes X')K_p(\ell)| = 2^p \prod_{i=1}^p x_{ii}^i. \text{ (Neudecker (1978)).} \end{aligned}$$

$$(9.2.18) \quad \text{If } Y = AXB + (AXB)' = AXB + B'X'A':$$

$$\begin{aligned} J(Y \rightarrow X) &= |K'_p(s)(B' \otimes A) + (A \otimes B')I_{(p,p)}K_p(\ell)| \\ &= |2K'_p(s)(B' \otimes A)K_p(\ell)| \\ &= 2^p \prod_{i=1}^p a_{ii}^i b_{ii}^{p-i+1} \text{ (Neudecker (1978)).} \end{aligned}$$

$$(9.2.19) \quad \text{If } Y = AXB' + (AXB')' = AXB' + BX'A':$$

$$\begin{aligned} J(Y \rightarrow X) &= |K'_p(s)(B \otimes A + (A \otimes B)I_{(p,p)})K_p(\ell)| \\ &= |2K'_p(s)(B \otimes A)K_p(\ell)| = 2^p \prod_{i=1}^p a_{ii}^i b_{ii}^{p-i+1} \text{ (Neudecker (1978)).} \end{aligned}$$

10. The maximum likelihood estimates of the parameters in the multivariate normal distribution: $X: p \times 1 \sim N(\underline{\mu}, \Sigma)$ based on a sample of size N .

To illustrate the three basic matrix derivative methods, consider the log likelihood function:

$$\ln L(\underline{\mu}, \Sigma) = -\frac{1}{2} N p \ln 2\pi + \frac{1}{2} N \ln |\Sigma^{-1}| - \frac{1}{2} \sum_{i=1}^N (\underline{x}_i - \underline{\mu})' \Sigma^{-1} (\underline{x}_i - \underline{\mu}).$$

Using the vector derivatives as in section 3:

$$\begin{aligned} \frac{\partial \ln L}{\partial \underline{\mu}'} &= -\frac{1}{2} \sum_{i=1}^N \left\{ \frac{\partial (\underline{x}_i - \underline{\mu})' \Sigma^{-1} (\underline{x}_i - \underline{\mu})}{\partial (\underline{x}_i - \underline{\mu})'} \cdot \frac{\partial (\underline{x}_i - \underline{\mu})}{\partial \underline{\mu}'} \right\}, \text{ from (3.2.3)} \\ &= -\frac{1}{2} \sum_{i=1}^N \{ 2(\underline{x}_i - \underline{\mu})' \Sigma^{-1} (-I_p) \}. \end{aligned}$$

Thus $\frac{\partial \ln L}{\partial \underline{\mu}} = \sum_{i=1}^N \Sigma^{-1} (\underline{x}_i - \underline{\mu})$ and since Σ^{-1} is nonsingular, $\frac{\partial \ln L}{\partial \underline{\mu}} = \underline{0}$ implies $\sum_{i=1}^N (\underline{x}_i - \underline{\hat{\mu}}) = \underline{0}$, $\underline{\hat{\mu}} = \underline{\bar{x}} = \frac{1}{N} \sum_{i=1}^N \underline{x}_i$

and $\sup_{\underline{\mu}} \ln L(\underline{\mu}, \Sigma) = -\frac{1}{2} N p \ln 2\pi + \frac{1}{2} N \ln |\Sigma^{-1}| - \frac{1}{2} \text{tr } \Sigma^{-1} A$,

where $A = \sum_{i=1}^N (\underline{x}_i - \underline{\bar{x}})(\underline{x}_i - \underline{\bar{x}})'$. Now consider the derivative with respect to Σ :

Using the element breadkwon method as in section 5.1 and 6.3:

$\frac{\partial \ln L}{\partial \sigma^{*ij}} = \frac{1}{2}(2 - \delta_{ij}) \left(\frac{\partial \ln L}{\partial \Sigma^{-1}} \right)_{ij}$, where Σ^{-1} is considered as mathematically independent and variable on the right hand side of the equation.

$$\begin{aligned} \frac{\partial \ln L}{\partial \sigma^{ij}} &= \frac{1}{2} N \left(\frac{\partial \ln |\Sigma^{-1}|}{\partial |\Sigma^{-1}|} \right)_{ij} - \frac{1}{2} \frac{\partial \text{tr } \Sigma^{-1} A}{\partial \sigma^{ij}} \\ &= \frac{1}{2} (N \Sigma^{-1} - A)_{ij}, \text{ from (2.1.1) and (2.1.2)} \end{aligned}$$

$\frac{\partial \ln L}{\partial \sigma^{*ij}} = 0$ implies $\frac{\partial \ln L}{\partial \sigma^{ij}} = 0$, and $\hat{\Sigma} = \frac{1}{N} A$, which is symmetric.

Using the Kronecker arrangement method:

Using the fact that:

$\frac{\partial \ln L}{\partial \Sigma^{-1}} = 0$ whenever $\frac{\partial \ln L}{\partial \Sigma^{-1}} = 0$, it follows that by considering Σ as mathematically independent and variable:

$$\begin{aligned} \frac{\partial \ln L}{\partial \Sigma^{-1}} &= \frac{\partial \left(\frac{N}{2} \ln |\Sigma^{-1}| \right)}{\partial \Sigma^{-1}} - \frac{\partial (\frac{1}{2} \text{tr } \Sigma^{-1} A)}{\partial \Sigma^{-1}} \\ &= \frac{N}{2} \frac{1}{|\Sigma^{-1}|} |\Sigma^{-1}| \Sigma - \frac{\partial (\frac{1}{2} \text{tr } \Sigma^{-1} A)}{\partial \Sigma^{-1}} = \frac{1}{2} (N \Sigma - A), \end{aligned}$$

$\frac{\partial \ln L}{\partial \Sigma^{-1}} = 0$ implies $\frac{\partial \ln L}{\partial \Sigma^{-1}} = 0$ and thus $\hat{\Sigma} = \frac{1}{N} A$.

Using the vector rearrangement method:

$$\begin{aligned}\frac{\partial \ln L}{\partial \text{vec}' \Sigma^{-1}} &= \frac{\partial \ln L}{\partial \text{vec}' \Sigma^{-1}} \frac{\partial \text{vec} \Sigma^{-1}}{\partial \text{vec}' \Sigma^{-1}} \\ &= \text{vec}' \left(\frac{\partial \ln L}{\partial \Sigma^{-1}} \right) K_p^{-1}(s) = \text{vec}' \left(\frac{1}{N} (N \Sigma - A) \right) K_p^{-1}(s).\end{aligned}$$

Thus $\frac{\partial \ln L}{\partial \text{vec}' \Sigma^{-1}} = 0$ implies $\text{vec}' \left(\frac{1}{N} (N \hat{\Sigma} - A) \right) = 0$, i.e. $\hat{\Sigma} = \frac{1}{N} A$.

11. CONCLUSION

The simplest and easiest to use of the three methods is the element breakdown method. This method lends itself as the natural method to obtain higher derivatives and is the most powerful when the gradient and Hessian of scalar loss functions with matrix argument are required.

The Kronecker arrangement method can be seen as an extension of the element breakdown method and is useful to obtain some second derivatives of scalar functions of matrices.

The vector derivatives and vector rearrangement matrix derivatives are particularly suitable to obtain Jacobians of matrix transformations and also higher derivatives. The chain rules associated with this method are closely related to the chain rules for scalar derivatives and are valid for scalar, vector and matrix valued functions of the argument matrix. Such flexibility unfortunately does not exist in the Kronecker arrangement derivatives. The vector rearrangement method also lends itself for the differentiation of functions with respect to the mathematically independent and variable elements of a patterned matrix, which can hardly be implemented in the Kronecker arrangement derivatives.

The element breakdown method combined with the vector and vector rearrangement method should enable the user to apply matrix derivatives with success in Statistics.

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